

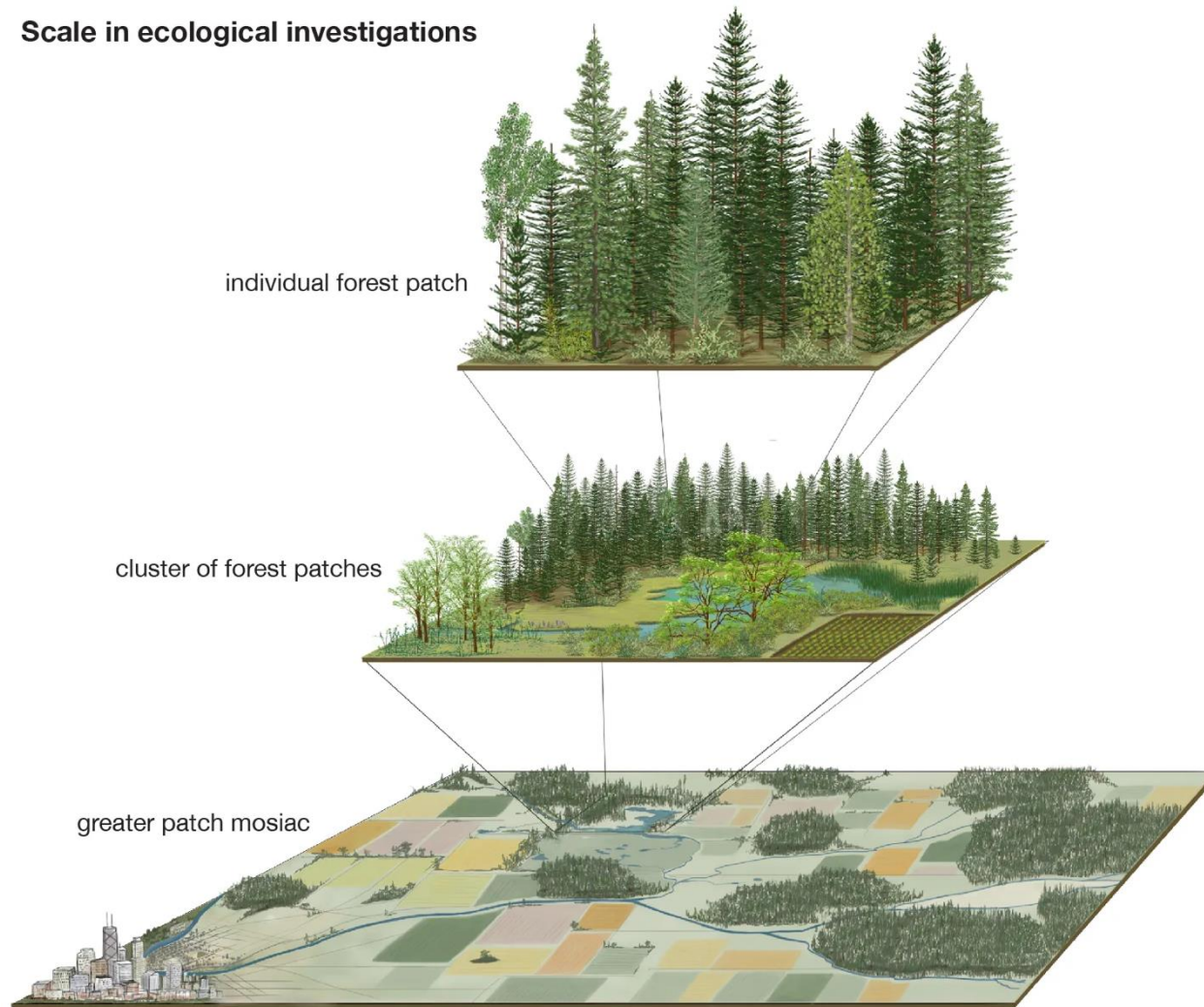
Analysing community data



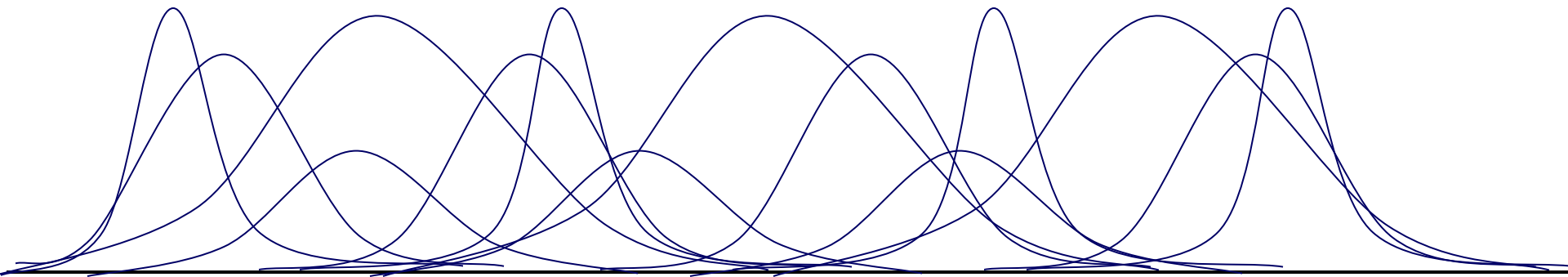
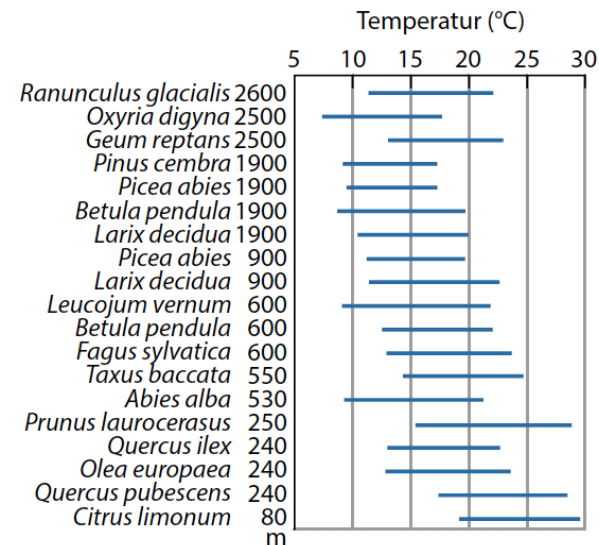
Tenerife, 3718m, Canary Islands, from Christoph Hormann, imagico.de



Scale in ecological investigations

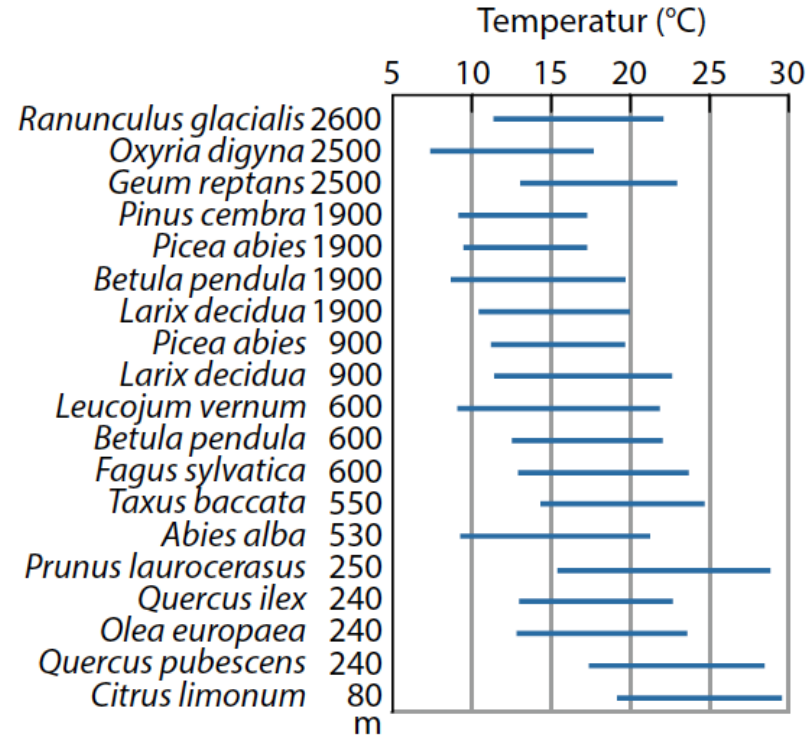
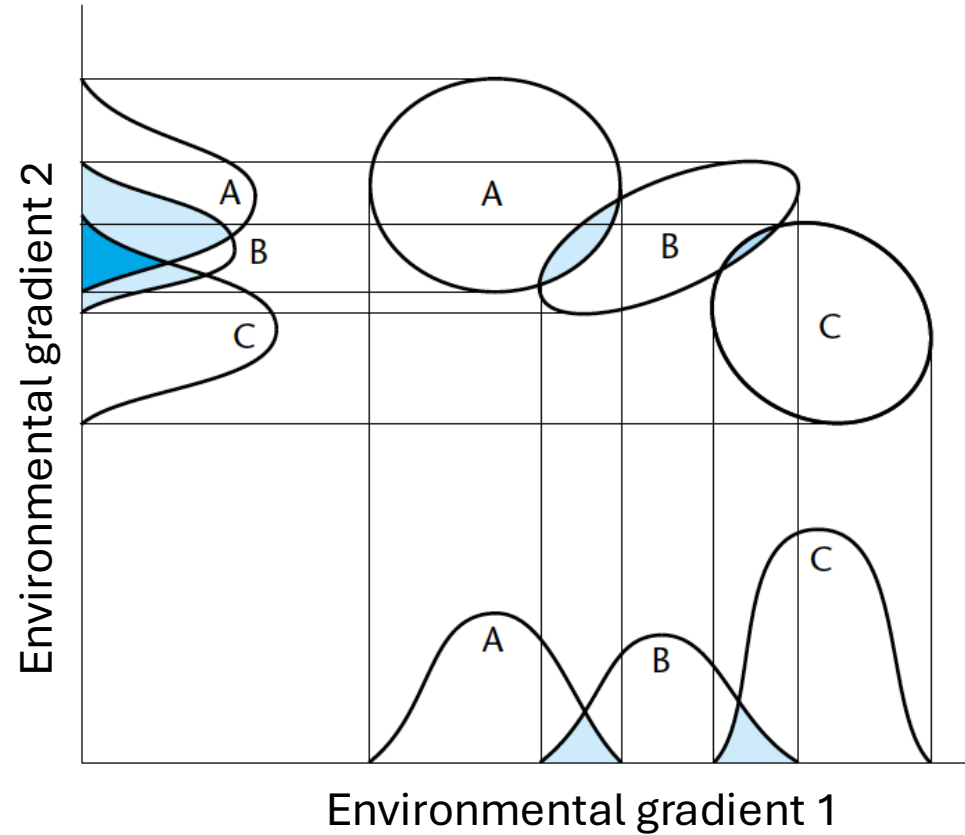


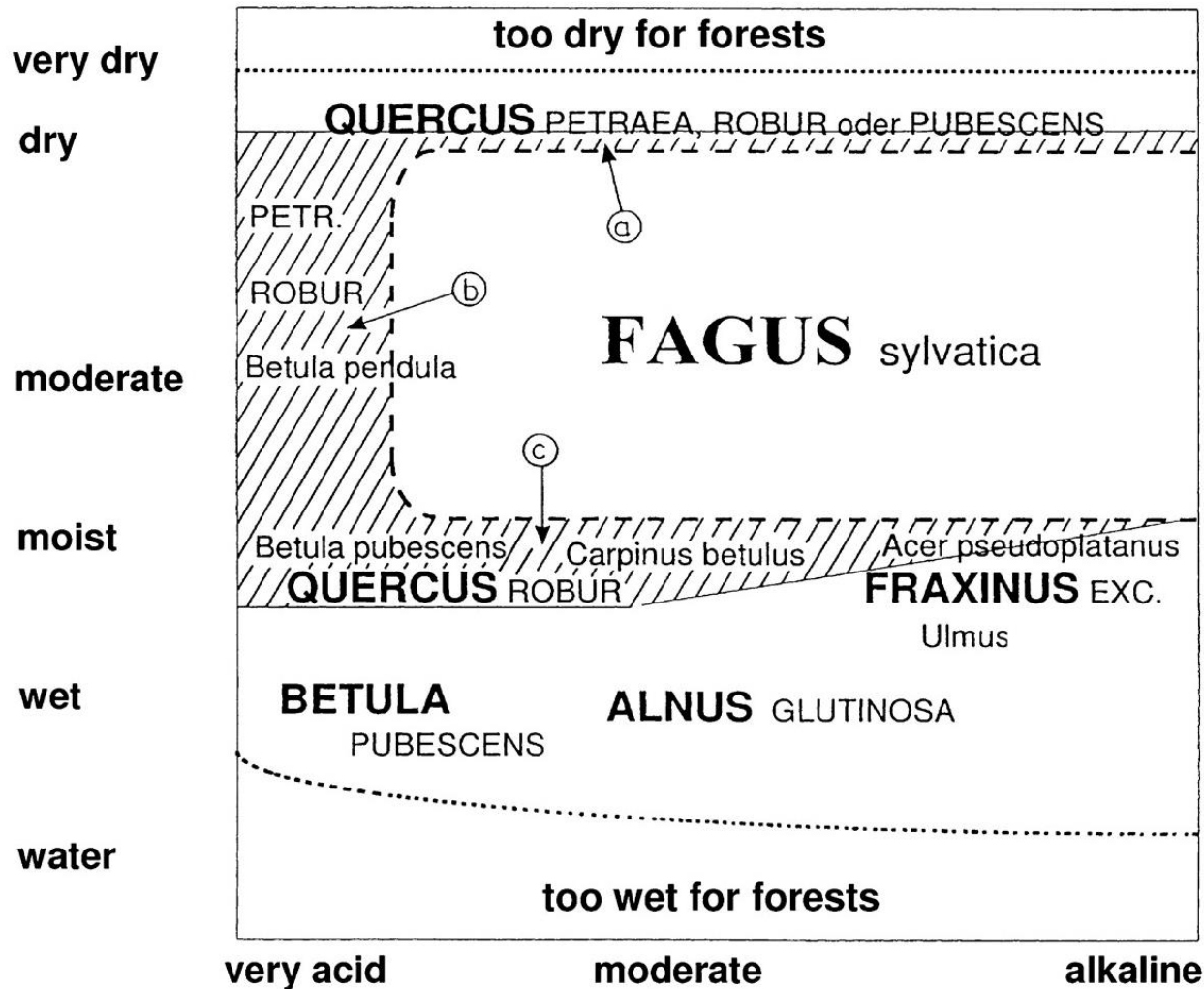
Environmental niche



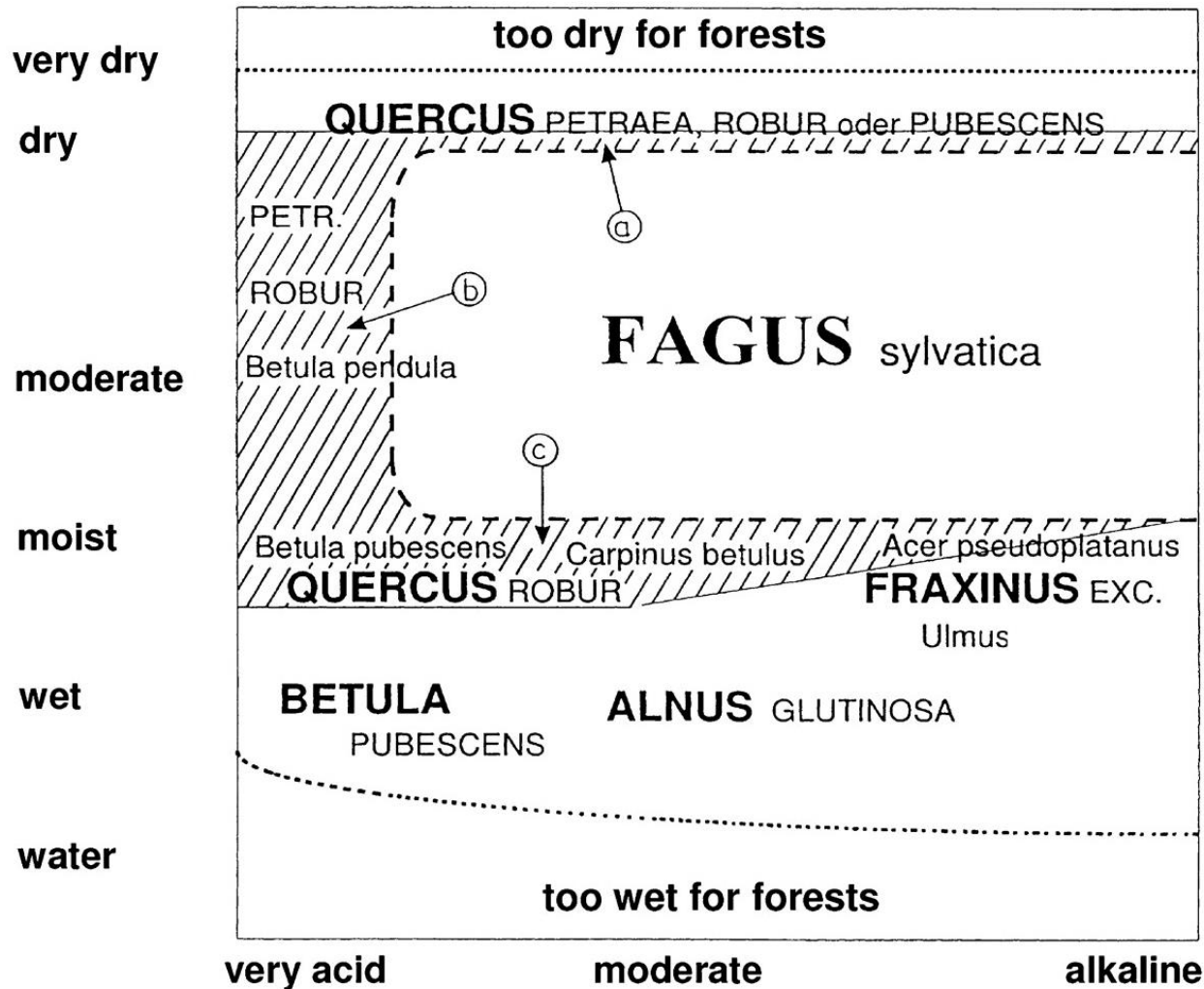
Temperature

Environmental niche

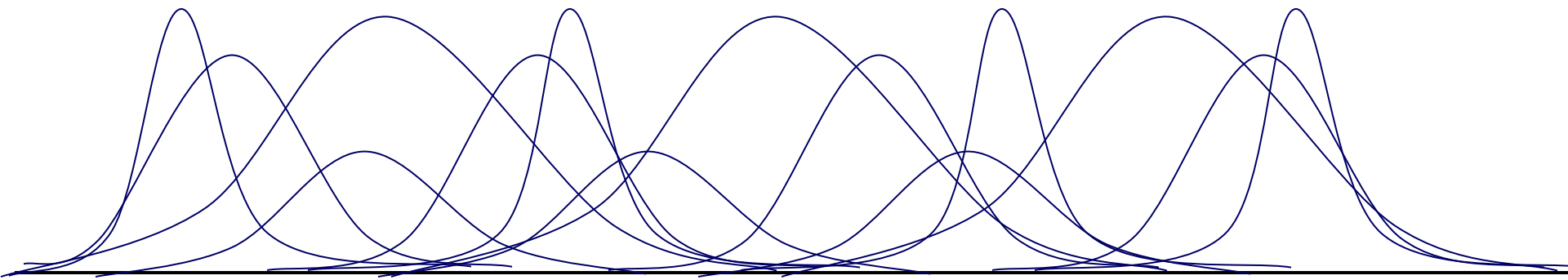




(from Ellenberg 2001)



(from Ellenberg 2001)



Environmental gradient

Sites

Community data

Species

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	
Achimill		4	4				1																	1						3				
Agrocani								4	5								6				1													
Agrocapi	3	4	2													8					1	1					4		5	3				
Agrorepe		1	1		7	5		4		4					9	3			3	5					5			9	5					
Agrostol					6			4	12									5				1			2	3						8		
Alopgeni								4	1									2						3		5					6			
Alloprat		1			7	7	1	2	5		1	1	4	2				4	5		3	2		5	1	1	3	2						
Anthodor	6						5						1				1					1					4			4				
Caltpalu				1													1	5					6											
Cardprat			2			2	1					1	3						1	1	2	1							1					
Caredist																		1			2													
Caregrac				4			5	8		1		7	2				4	3					1	2		2							1	
Careprae		2	2		1	1										2		1							2		1	3	1					
Carevesi				4			1	4									1	7				2												
Carevulp								1		1		1	1					3	1					1										
Cirsarve							1							4						3											1			
Cirslanc																							1											
Cniddubi																														2				
Desccesp																8		1		5	4									3	1			
Eleounig										1	2									1	1													
Eropvern	1					1																					1	1						
Euphesul			1																										1		1			
Festprat							2	1			1	1	3	2																1				
Galipalu				1			2	1		1	1	3	2					3		1		1	3	1	1	2						1		
GaliverA	1	1	3											4													4		1	4				
Glychui							1												2												2			
Glycmayi								1			2																					1	11	10
Holclana						4	2						4			1			2	6											1			
Junceffu			1		1	1								1							4	1			1									
Lathprat												1	1																		1			
Lotucom		1	1		1																	1									1			
Lychflos						2																1		1										
Phalarun						5		1	2	2	1	11	8	2			1		1	2	1	1	6			1						1		
Planinte										3	3	1									1					1								
Poa palu				1		1	1		11		4		1	1			1			6	2		4	1		6	3							
Poa praA		3	2	1	5	6	1							3		1				1		4	1	2	2			4	2					
Poa triv				1		2	1	1	4												2		2	3		3	1							
Poteanse				1		2										1			1		2													
Poterept								1																		1								
Ranuflam				1			1	3					1								1						3							
Ranurepe		4	1	1	1	2	4	1	4		1	3					3		5	4	4	2	6		9	9								
Roriamph									4		7																1							
RorisylA									1	2	1																							
Rumeacet					1		4								4																			
Rumethyr	1	5	2		3	5											3																	
Siumlati						1			2				1										2	1										
Stelpalu				2		1					1	1	1					1	1	2			1	1										
Sympoffi																																		
TaraoffA			1	1	1	1						1					1			1	1		1		2				1	1				
Trifrepe							3																											
Vicicrac				1		1				9		4																			1	2		
Vicilath		1																																
Vicitefr		1	1											2													1	1						

Species

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	
Achmill		4	4				1																		1						3			
Agrocani								4	5								6				1													
Agrocapi		3	4	2												8					1	1					4			5	3			
Agrorepe			1	1		7	5			4		4			9	3			3	5						5			9	5				
Agrostol							6			4	12								5			1				2	3							
Aloppengi										4	1								2						3		5					8	6	
Alopprat			1			7	7	1				5		1	1	4	2			4	5		3	2		5	1	1	3	2				
Anthodor		6						5						1				1				1						4			4			
Caltpalu					1												1	5					6											
Cardprat				2			2	1					1	3					1	1	2	1							1					
Caredist																			1		2													
Caregrac				4			5	8		1			7	2			4	3				1	2			2							1	
Careprae		2	2		1	1										2			1						2			1	3	1				
Carevesi				4			1	4									1	7					2											
Carevulp									1		1		1	1					3		1				1									
Cirsarve								1						4							3											1		
Cirslanc																							1											
Cniddubi																																		
Desccesp															8		1			5		4								2	3	1		
Eleounig										1		2							1		1													
Eropvern	1						1																					1	1					
Euphesul			1																										1		1			
Festprat								2	1				1	1	3	2																	1	
Galipalu				1					2	1		1	1	3	2				3		1		1	3	1	1	2							
GaliverA	1	1	3												4													4			1	4		
Glycfliu								1											2							7						2		
Glycmaxi									1			2																				1	11	10
Holclana							4	2					4				1			2	6		4	1						1				
Junceffu				1			1	1																4	1		1							
Lathprat													1	1																				
Lotucorn			1	1			1															1										1		
Lychfilos								2														1		1										
Phalarun				5			1	2		2	1	11	8	2				1		1	2	1	1	6								1		
Planinte									3	3	1								1							1								
Poa palu							1	1		11		4		1			1			6	2		1				6	3						
Poa praA		3	2		1	5	6	1						3		1			1		4		2			2			4	2				
Poa triv				1		2	1	1	4												2		2	3		3	1							
Poteanse				1		2										1																		
Poterept									1																		1							
Ranufflam				1			1	3					1								1							3						
Ranurepe		4	1	1	1	2		4	1	4			1	3			3		5	4	4	2	6			9	9							
Roriamp									4		7																							
RorisylA								1	2	1						1									3		4							
Rumeacet				1			4								4																		1	
Rumethyr		1	5	2		3	5										3																1	
Siumlati				1				2				1											2	1										
Stelpalu				2			1				1	1	1				1	1	2			1	1											
Sympoffi																																		
TaraoffA				1	1	1						1							1	1		1				2				1	1			
Trifrepe				2			3																											
Vicicrac				1			1				9			4																	1	2		
Vicilath																																		
Vicitr		1	1												2													1	1					



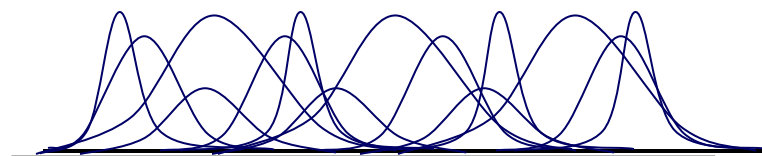
Species

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Vicilath	1																																	
Achimill	4									1					1																			
Agrocapi	3	4	4	2	8	3	5						1							1														
Euphesul					1		1	1																										
GaliverA	1	1	4	3		4	1					4																						
Eropvern	1		1						1		1																							
Rumethyr	1	5	4	2	3	1			3		5				2																			
Vicitetr	1	1	1					1				2																						
Cniddubi						2																												
Careprae	2	1	2	2		1	3	1		1				2			1																	
Anthodorr	6		4		4								1	1		5					1													
Lotucorn				1		1						1	1						1															
Poa praA	3		2	1		2	4	5		6	3	4		2	1				1	1						2								
Agrorepe	1		1	3		5	9	7	9	5				5	5		4	3					4											
TaraoffA				1	1	1	1		1		1			1	2		1	1	1	1														
Cirsarve					1						4		3		1																			
Rumeacet		1			1						4	4			4			1																
Lathprat					1						1		1										1											
Alopprat	1	1		2		2	3	7	4	7	1		5	5	1	5	4		3		3	2	1		2		1							
Desccesp					1	3					8										4	1												
Vicirac					2	1					4					1	9		1															
Holclana					1						4	6	2		4						1							2						
Festprat						1									2												1							
Cardprat						1					3	2	1		2			1	2	1			1				1							
Poteanse											1		2						1															
Lychflos						1	1				1		1		2									1										
Trifrepe												1	4		3		4	2	1															
Caredist						2																		1										
Junceffu						1	4				1			1			1	1									1	1						
Carevulp						1	1				1		1	1	3							1	1											
Poterept														1			1					1												
Stelpalu											1	1		1	1	2	2	1	1					1	1									
Poa triv									2				2	3	1			1	2	4						3	1	1						
Poa palu							1	1					2	6	1	4	6	1	4	1	1	1		1		1	3							
Eleounig										1				1		1						2												
Ranurepe									1	1	3	4	4	9	1	4	5	4	2	3	4	1			6	2	9	1						
RorisylA								1						4		1	1					1		3				2						
Agrocani											1				4							6			5									
Cirsilanc																				1														
Sympoffi																			3						1									
Caltpalu																				1	6	1			5									
Planinte																1	1	1				3						3						
Carevesi														1				4	2	1					7	4								
Ranufлам													1					1					1			3	3							
Caregrac											2			5			4	1	4				7	3	2	8	2	1				1		
Phalarun												1	2		1	1	1	5	1	1		2	2		6	2	1	1	1	8				
Siumlati																		1	2	3					1	2	1			1				
Agrostol											6				2			5		1		4						3	12	8				
Galipalu														2				1		1		1	3	3	3	2				1	1		1	
Alopgeni																	2					4												
Roriamph																																		
Glycflui																																		
Glycmaxi																																		

Sites

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	
Achmill		4	4				1																		1						3			
Agrocani								4	5									6			1													
Agrocapi	3	4	2													8					1	1					4		5	3				
Agrorepe		1	1		7	5				4	4					9	3			3	5					5			9	5				
Agrostol						6				4	12								5			1				2	3						8	
Alopgeni										4	1								2					3		5		1	1	3	2		6	
Alopprat		1			7	7	1			2	5		1	1	4	2			4	5		3	2			5	1	1	3	2				
Anthodor	6						5						1				1					1						4				4		
Caltpalu				1													1	5					6											
Cardprat			2				2	1					1	3					1	1	2	1								1				
Caredist																		1			2													
Caregrac			4				5	8		1		7	2					4	3				1	2		2							1	
Careprae	2	2			1	1										2		1							2			1	3	1				
Carevesi			4				1	4									1	7				2												
Carevulp									1		1		1	1				3		1					1									
Cirsarve							1						4						3												1			
Cirslanc																							1											
Cniddubi																															2			
Desccesp														8			1		5		4									3	1			
Eleounig											1	2							1		1													
Eropvern	1					1																						1	1					
Euphesul			1																									1		1				
Festprat							2	1																						1				
Galipalu				1			2	1		1	1	3	2					3		1		1	3	1	1	2							1	
GaliverA	1	1	3										4														4		1	4				
Glycfui							1												2												2			
Glycmaxi								1			2												1	2	3		1				1	11	10	
Holciana						4	2						4			1			2	6											1			
Junceffu			1			1	1						1							4	1					1								
Lathprat												1	1																			1		
Lotucorn		1	1		1																1											1		
Lychfios			1				2													1		1	1											
Phalarun			5			1	2	1	2	2	1	11	8	2			1		1	2	1	1	6			1								
Planinte									3	3	1								1							1								
Poa palu			1		1	1			11		4		1		1		1		6	2		4	1			6	3							
Poa praA	3	2	1	5	6	1							3		1				1	4		2				2				4	2			
Poa triv			1		2	1	1	4												2		2	3			3	1							
Poteanse			1		2									1				1		2														
Poterept								1											1							1								
Ranufiam			1			1	3						1							1							3							
Ranurepe			4	1	1	1	2	4	1	4			1	3			3		5	4	4	2	6			9	9							
Roriamp									4		7														1		4	1						
RorisyA								1	2	1						1								3		4								
Rumeacet				1			4							4									4					1				1		
Rumethyr	1	5	2		3	5										3										2		4				1		
Siumlati				1				2				1					3						2	1			1							
Stelpalu			2				1				1	1	1					1	1	2		1	1											
Sympoffi																									3	1								
TaraoffA			1	1	1	1					1					1			1	1		1				2					1	1		
Trifrepe			2				3																											
Vicicrac				1			1				9		4																			1	2	
Viciliath		1																																
Vicietr		1	1										2															1	1					

Species

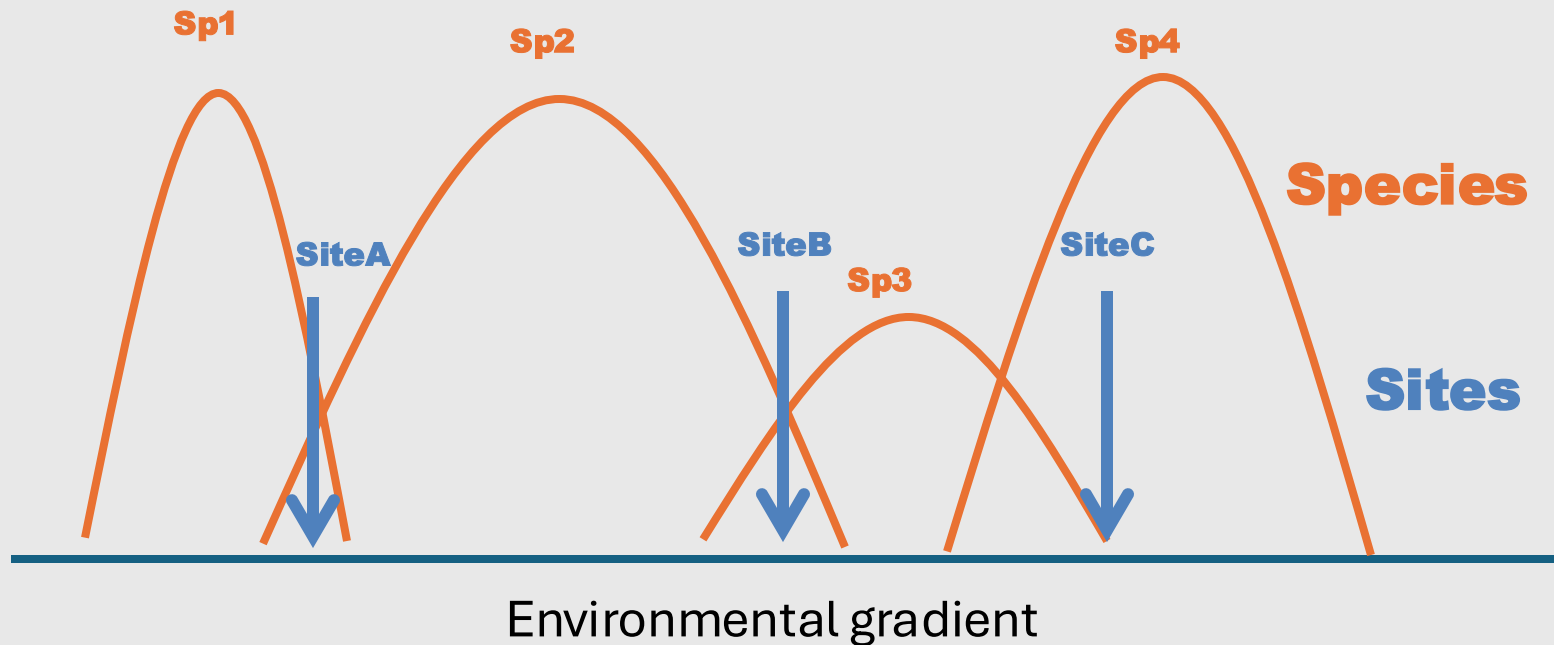
[illegible]

Species



Ordination - Correspondence Analysis

What is it trying to do?



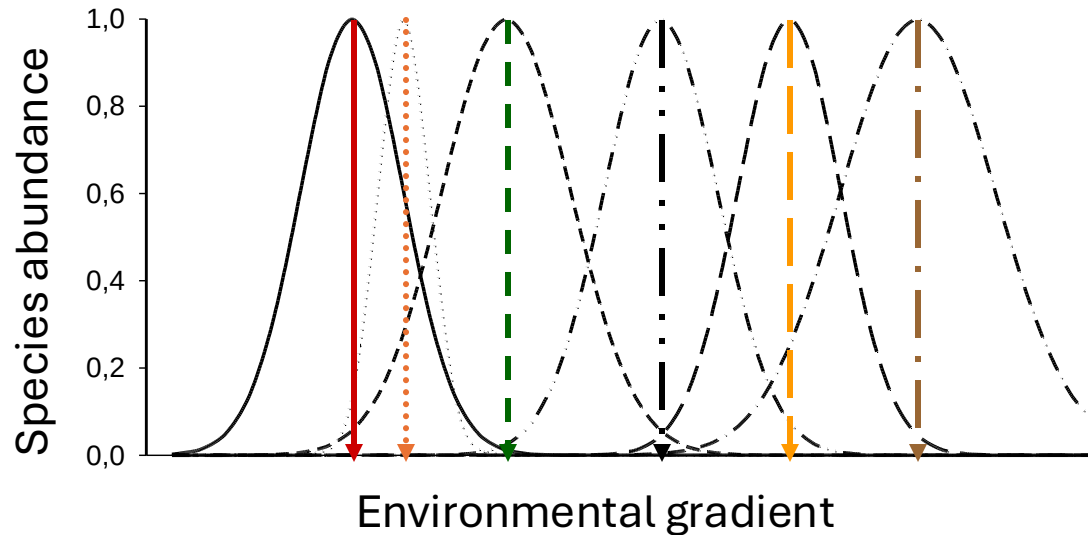
If you knew the species or the sites you could position the other

Correspondence Analysis

Assumes a **unimodal** relationship of species along important gradients

The principle:

Estimates the **optimum** of a species using a **weighted average**



Sites

Species

	1	2	3	16	30	29	28	5	15	6	14	21	20	4	22	17	9	13	18	23	8	26	10	31	12	24	33	32
Vicilath	1																											
Achmill	4	4	3					1		1																		
Agrocapi	3	4	4	2	8	3	5			1					1													
Euphesul			1		1	1																						
GaliverA	1	1	4	3	4	1				4																		
Eropvern	1		1				1			1																		
Rumethyr	1	5	4	2	3	1		3	5			2																
Vicitehr		1	1	1				1		2																		
Cniddubi							2																					
Careprae	2	1	2	2		1	3	1		1			2		1													
Anthodor	6		4		4					1	1			5			1											
Lotucorn			1	1						1	1				1													
Poa praA	3	2	1		2	4	5		6	3	4		2	1	1	1					2							
Agrorepe	1		1	3	5	9	7	9	5			5	5	4	3			4										
TaraoffA			1	1	1	1	1	1	1		1	2	1	1	1	1												
Cirsarve				1						4	3	1																
Rumeacet		1		1						4	4		4		1													
Lathprat				1						1	1								1									
Alopprat	1	1		2	2	3	7	4	7	1	5	5	1	5	4	3	2	1		2	1							
Descocesp				1	3					8	5					4	1											
Vicicrac				2	1					4			1	9	1													
Holdlana				1						4	6	2	4			1					2							
Festprat					1								2								1							
Cardprat				1						3	2	1	2	1	2	1			1		1							
Poteanse										1	2	2		1					1									
Lychflos										1	1	2		1						1								
Trifrepe										1	4	3	4	2	1													
Caredist										2									1									
Junceffu										1	4		1		1	1					1	1						
Carevulp										1	1		1	3			1	1										
Poterept											1					1												
Stelpalu										1	1		1	1	2	2	1	1	1	1								
Poa triv										2		2	3	1		1	2	4			3	1	1					
Poa palu										1	1		2	6	1	4	6	1	4	1	1	1	3					
Eleounig											1								2									
Ranurepe										1	1	3	4	4	9	1	4	5	4	2	3	4	1	6	2	9	1	
RorisyA										1			4	1	1			1			3		2					
Agrocani											1		4					6			5							
Cirslanc															1													
Sympoffi															3					1								
Caltpalu															1	6	1		5									
Planinte											1	1	1				3						3					
Carevesi												1			4	2	1		7	4								
Ranufiam											1				1				1		3	3						
Caregrac											2		5		4	1	4	7	3	2	8	2	1			1		
Phalarun												1	2	1	1	1	5	1	1	2	2	6	2	1	1	1	8	
Siumlati																	1	2	3			1	2	1				
Agrostol											6		2		5		1	4			3	12	8					
Galipalu											2	1	1	1	1	1	1	3	3	3	2			1	1	1		
Aloppeni															2			4			5	1	6	3				
Roriamph																					1	4		7	1			
Glycflui															2							2	7					
Glycmaxi																1					2	1	1	1	2	3	10	11

Correspondence Analysis

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	GM		
Vicilath		1																															-1.441		
Achimill		4	4		3			1																									-1.218		
Agrocapi	3	4	4	2	8	3	5				1											1											-1.217		
Euphesul				1	1	1																											-1.194		
GaliverA		1	1	4	3	4	1				4																						-1.141		
Eropvern		1	1																														-1.117		
Rumethyr	1	5	4	2	3	1					3	5									2												-1.083		
Viclatr		1	1	1											2																		-1.027		
Cnidubi							2																										-0.967		
Careprae		2	1	2	2		1	3	1		1										2		1										-0.955		
Anthodor	6		4								1	1									5			1									-0.946		
Lotucom					1						1	1																					-0.653		
Poa praA		3	2	1			2	4	5		6	3	4								2	1		1									-0.633		
Agrorepe		1	1	3		5	9	7	9	5					5	5					4	3											-0.574		
TaraoffA				1	1	1					1				1	2		1	1	1	1												-0.492		
Cinsarve											4	3	1																					-0.440	
Rumeacet			1								4	4									4													-0.400	
Lathprat											1	1																						-0.391	
Alopprat		1	1		2		2	3	7	4	7	1			5	5	1	5	4		3		2	1										-0.378	
Descocesp											8	5										4	1											-0.357	
Vicicrac											4										1	9	1											-0.287	
Holclana											4	6	2																					-0.219	
Festprat																																		-0.186	
Cardprat											3	2	1								2	1	2	1										-0.117	
Poteanse											1	2	2																					-0.091	
Lychfios											1	1																						-0.059	
Trifrepe												1	4	3							4	2	1											-0.034	
Caredist												2																						-0.015	
Junceffu											1	4									1	1												-0.001	
Carevulp											1	1									1	3												0.021	
Poterept																																			0.072
Stelpalu												1	1								1	2	2	1	1									0.097	
Poa triv											2																							0.099	
Poa palu											1	1																						0.117	
Eleounig																																			0.124
Ranurepe											1	1	3	4	4	9	1	4	5	4	2	3	4	1										0.131	
RorisylA											1				4	1	1																	0.164	
Agrocapi																																			0.217
Cinslanc																																			0.225
Sympoffi																																			0.295
Caltpalu																																			0.322
Planinte																																			0.361
Carevesi																																			0.363
Ranufiam																																			0.383
Caregrac																																			0.399
Phalarun																																			0.453
Siumlati																																			0.476
Agrostol																																			0.512
Gallpalu																																			0.577
Aloppeni																																			1.009
Roriamp																																			1.298
Glyclui																																			1.753
Glycmari																																			3.117

The principle:

Calculate the **weighted average of species** from **site** values (current order of sites)

$$\tilde{x}_j = \frac{\sum_{i=1}^n x_i y_{ij}}{\sum_{i=1}^n y_{ij}}$$

x_j = **Weighted average of species j**

n = **Number of sites**

y_{ij} = **Abundance of species j in site i**

x_i = **Site score of site i**

CA – How it Works

Raw data

samples	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
<i>Achmil</i>	1	3			2	2	2			4							2			
<i>Agrsto</i>			4	8				4	3			4	5	4	4	7			5	
<i>Airpra</i>																	2		3	
<i>Alogen</i>		2	7	2				5	3			8	5			4				
<i>Antodo</i>					4	3	2			4							4		4	
<i>Belper</i>		3	2	2	2					2								2		
<i>Brohor</i>		4		3	2		2			4										
<i>Chealb</i>														1						
<i>Cirarv</i>				2																
<i>Elepal</i>								4						4	5	8				4
<i>Elyrep</i>	4	4	4	4	4				6											
<i>Emprig</i>																			2	
<i>Hyprad</i>										2							2		5	
<i>Junart</i>								4	4						3	3				4
<i>Junbuf</i>							2		4			4	3							
<i>Leoaut</i>		5	2	2	3	3	3	3	2	3	5	2	2	2	2		2	5	6	2
<i>Lolper</i>	7	5	6	5	2	6	6	4	2	6	7							2		
<i>Plalan</i>					5	5	5			3	3						2	3		
<i>Poapra</i>	4	4	5	4	2	3	4	4	4	4	4		2				1	3		
<i>Poatri</i>	2	7	6	5	6	4	5	4	5	4		4	9			2				
<i>Potpal</i>															2	2				
<i>Ranfla</i>								2					2	2	2	2				4
<i>Rumace</i>					5	6	3		2			2								
<i>Sagpro</i>				5				2	2		2	4	2						3	
<i>Sairep</i>																		3	3	5
<i>Tripa</i>					2	5	2													
<i>Trirep</i>		5	2	1	2	5	2	2	3	6	3	3	2	6	1			2	2	
<i>Viclat</i>										1	2							1		
<i>Brarut</i>			2	2	2	6	2	2	2	2	4	4			4	4		6	3	4
<i>Calcus</i>														4		3				3
<i>Hiprha</i>									1											
<i>Poaann</i>	3	3		4			2		2		2							2	1	
<i>Ranacr</i>						3	2		2	3			3	1	1			4		

CA – How it Works

Calculating weighted mean of **species**

samples	1	2	5	6	7	11	18	3	4	10	17	9	12	8	13	14	15	16	19	20
Achmil	1	3	2	2	2					4	2									
Agrsto								4	8			3	4	4	5	4	4	7	5	
Airpra											2									
Al ogen		2						7	2			3	8	5	5			4		
Antodo			4	3	2					4	4								4	
Belper		3	2				2	2	2	2										
Brohor		4	2		2				3	4										
Chealb															1					
Cirarv									2											
Elepal														4		4	5	8		4
Elyrep	4	4	4					4	4			6								
Emprnig																			2	
Hyprad						2					2								5	
Junart												4		4			3	3		4
Junbuf															3					
Leoaut															2	2	2		6	2
Lolper															1					
Plalan																				
Poapra	4	4													1	2				
Poatri	2	1													1	9		2		
Potpal																	2	2		
Ranfla															2	2	2	2		4
Rumace			5	6	3							2	2							
Sagpro						2			5			2	4	2	2				3	
Sairep							3												3	5
Tripra			2	5	2															
Trirep		5	2	5	2	3	2	2	1	6		3	3	2	2	6	1		2	
Vi clat						2	1			1										
Brarut			2	6	2	4	6	2	2	2		2	4	2			4	4	3	4
Calcus																4		3		3
Hiprha							2					1							1	
Poaann	3	3			2	2	4		4			2								
Ranacr				3	2					3		2			3	1	1			

1. Iteration

Site score is random =
e.g. order of plots

$$\tilde{x}_j = \frac{\sum_{i=1}^n x_i y_{ij}}{\sum_{i=1}^n y_{ij}}$$

CA – How it Works

Sort species by weighted species scores

samples	1	2	5	6	7	11	18	3	4	10	17	9	12	8	13	14	15	16	19	20	
<i>Cirarv</i>									2												0,4
<i>Chealb</i>															1						0,7
<i>Emprnig</i>																			2		1,9
<i>Viola</i>							2	1		1											2,5
<i>Tripra</i>			2	5	2																2,7
<i>Potpal</i>																2	2				2,9
<i>Hiprha</i>								2		3	4		1						1		3,2
<i>Brohor</i>		4	2		2																4,2
<i>Belper</i>		3	2				2	2	2	2											4,3
<i>Airpra</i>												2							3		4,6
<i>Elyrep</i>	4	4	4					4	4			6									5,7
<i>Achmil</i>	1	3	2	2	2					4	2										5,9
<i>Rumace</i>			5	6	3							2	2								6,2
<i>Junbuf</i>					2							4	4								6,9
<i>Ranacr</i>				3	2					3		2			3	1	1				7,4
<i>Hyprad</i>						2					2								5		7,6
<i>Poaann</i>	3	3			2	2	4		4			2									7,6
<i>Calcus</i>																4		3	3		8,2
<i>Sagpro</i>							2		5			2	4	2	2				3		10
<i>Saltrep</i>							3												3	5	11
<i>Ranfla</i>														2	2	2	2	2	4		11
<i>Antodo</i>			4	3	2					4	4								4		12
<i>Junart</i>												4		4			3	3	4		12
<i>Plalan</i>			5	5	5	3	3			3	2										12
<i>Alogen</i>		2						7	2			3	8	5	5			4			16
<i>Poapra</i>	4	4	2	3	4	4	3	5	4	4	1	4		4	2						17
<i>Lolper</i>	7	5	2	6	6	7	2	6	5	6		2		4							18
<i>Elepal</i>													4			4	5	8		4	19
<i>Trirep</i>		5	2	5	2	3	2	2	1	6		3	3	2	2	6	1		2		22
<i>Poatri</i>	2	7	6	4	5			6	5	4		5	4	4	9			2			23
<i>Agrsto</i>								4	8			3	4	4	5	4	4	7	5		27
<i>Brarut</i>			2	6	2	4	6	2	2	2		2	4	2		4	4	3	4		29
<i>Leoaut</i>		5	3	3	3	5	5	2	2	3	2	2	2	3	2	2	2	6	2		30

CA – How it Works

Calculating weighted mean of **sites**

samples	1	2	5	6	7	11	18	3	4	10	17	9	12	8	13	14	15	16	19	20
Clrarv									2											0,4
Chealb															1					0,7
Emprnig																			2	1,9
Violat							2	1			1									2,5
Tripra			2	5	2															2,7
Potpal																2	2			2,9
Hiprha								2		3	4		1						1	3,2
Brohor		4	2		2															4,2
Belper		3	2				2	2	2	2										4,3
Airpra												2							3	4,6
Elyrep	4	4	4					4	4			6								5,7
Achmil	1	3	2	2	2					4	2									5,9
Rumace			5	6	3							2	2							6,2
Junbuf					2							4	4							6,9
Ranacr				3	2						3	2			3	1	1			7,4
Hyprad						2					2							5		7,6
Poaann	3	3			2	2	4		4			2								7,6
Calcus																4		3	3	8,2
Sagpro							2		5			2	4	2	2				3	10
Saltrep							3												3	11
Ranfla															2	2	2	2	2	11
Antodo			4	3	2					4	4								4	12
Junart												4		4			3	3	4	12
Plalan			5	5	5	3	3			3	2									12
Alogen		2							7	2		3	8	5	5			4		16
Poapra	4	4	2	3	4	4	3	5	4	4	1	4		4	2					17
Lolper	7	5	2	6	6	7	2	6	5	6		2		4						18
Elepal														4		4	5	8		19
Trirep		5	2	5	2	3	2	2	1	6		3	3	2	2	6	1		2	22
Poatri	2	7	6	4	5			6	5	4		5	4	4	9			2		23
Agrsto								4	8			3	4	4	5	4	4	7	5	27
Brarut			2	6	2	4	6	2	2	2		2	4	2		4	4	3	4	29
Leoaut		5	3	3	3	5	5	2	2	3	2	2	2	3	2	2	2	6	2	30

1,6 2,8 2,7 2,6 2,7 1,7 2,4 1,0 3,8 2,2 1,1 3,9 2,5 0,6 1,9 1,4 0,4 0,7 3,7 2,3

$$\tilde{x}_i = \frac{\sum_{j=1}^n \tilde{x}_j y_{ij}}{\sum_{j=1}^n y_{ij}}$$

CA – How it Works

Sort by weighted site scores

samples	15	8	16	3	17	14	1	11	13	10	20	18	12	6	7	5	2	19	4	9		
Cirarv																			2		0,2	
Chealb									1												0,3	
Emprnig																	2				0,3	
Violat								2		1		1									0,5	
Tripra														5	2	2					0,5	
Potpal	2					2															0,6	
Hiprha												2						1		1	1	
Brohor									4						2	2	4		3		1	
Belper				2					2		2					2	3		2		1	
Airpra					2													3			1,1	
Elyrep				4				4								4	4		4	6	1,1	
Achmil					2			1		4				2	2	2	3				1,5	
Rumace													2	6	3	5				2	1,5	
Junbuf									3						2					4	1,6	
Ranacr	1					1			3	3				4		3	2			2	1,6	
Hyprad					2			2											5		2,2	
Poaann							3	2				4			2		3		4	2	2,2	
Calcus			3			4					3										2,3	
Sagpro		2						2	2				4					3	5	2	2,5	
Saltrep											5	3									2,5	
Ranfla	2	2	2			2			2		4							3			2,6	
Antodo					4					4				3	2	4		4			3,5	
Junart	3	4	3								4									4	3,6	
Plalan					2			3		3		3		5	5	5					4,3	
Alogen		5	4	7					5				8				2		2	3	4,9	
Poapra		4		5	1		4	4	2	4		3		3	4	2	4		4	4	4,9	
Lolper		4		6			7	7		6		2		6	6	2	5		5	2	6,3	
Elepal	5	4	8			4					4										6,3	
Trirep	1	2			2		6		3	2	6		2	3	5	2	2	5	2	1	3	6,7
Poatri		4	2	6			2		9	4				4	4	5	6	7		5	5	7,1
Agrsto	4	4	7	4		4			5		5			4					8	3	7,2	
Brarut	4	2	4	2				4		2	4	6	4	6	2	2		3	2	2	8,3	
Leoaut	2	3		2	2	2		5	2	3	2	5	2	3	3	3	5	6	2	2	9,9	
	0,4	0,6	0,7	1,0	1,1	1,4	1,6	1,7	1,9	2,2	2,3	2,4	2,5	2,6	2,7	2,7	2,8	3,7	3,8	3,9		

CA – How it Works

Further iterations until no changes

samples	17	15	14	20	19	16	1	18	12	11	13	5	8	7	9	10	6	4	2	3	
Emprnig					2																0,5
Chealb											1										0,8
Potpal		2	2																		1
Airpra	2				3																1,1
Cirarv																		2			1,4
Hiprtha					1			2							1						1,7
Viclat								1		2						1					2,1
Hyprad	2				5					2											2,7
Calcus			4	3	3																2,9
Salrep				5	3			3													3,4
Ranfla																					-
Tripra																					
Junbuf																					
Junart																					
Belper																					
Elepal																					
Ranacr																					
Achmil																					
Antodo																					
Brohor																					
Sagpro																					
Rumace																					
Poaann																					
Plalan	2						3	4		2											11
Elyrep								3		3		5		5		3	5	4	3		15
Alogen																					16
Agrsto		4	4	5		7		4		5		4		3		8		4			21
Brarut		4		4	3	4		6	4	4		2	2	2	2	2	6	2			24
Trirep		1	6		2			2	3	3	2	2	2	3	6	5		5	2		26
Leaaut	2	2	2	2	6			5	2	5	2	3	3	3	2	3	3	2	5	2	28
Poapra	1							4	3		4	2	2	4	4	4	3	4	4		29
Lolper								7	2		7		2	4	6	2	6	6	5	5	36
Poatri								2	2		4		9	6	4	5	5	4	4	5	39
	5,4	11	11	12	12	16	17	19	23	24	25	26	27	29	29	30	32	32	32	32	

Definition of first axis: further axes are calculated on the same principle but for remaining unexplained variance

Correspondence Analysis

	1	2	3	16	30	29	28	5	15	6	14	21	20	25	7	11	19	4	22	17	9	13	18	23	8	26	10	31	12	14	24	33	32	GM			
Vicilath		1																																-1.441			
Achimill			4	4	3					1				1																					-1.218		
Agrocapi	3	4	4	2	8	3	5					1						1																	-1.217		
Euphesul				1	1	1	1												1																-1.194		
GaliverA	1	1	4	3	4	1					4																								-1.141		
Eropvern	1	1					1			1																										-1.117	
Rumethyr	1	5	4	2	3	1			3	5				2																						-1.083	
Vicitefr		1	1	1				1			2																									-1.027	
Cnidubi					2																															-0.967	
Careprae	2	1	2	2	1	3	1			1			2		1																				-0.955		
Anthodor	6		4		4					1	1			5				1																		-0.946	
Lotucorn				1	1					1	1						1																			-0.653	
Poa praA	3	2	1		2	4	5		6	3	4		2	1		1								2												-0.633	
Agrorepe	1	1	3	5	9	7	9	5					5	5	4	3					4															-0.574	
TaraofA		1	1	1	1	1			1		1		2	1	1	1	1																			-0.492	
Cirsarve				1							4	3	1																								-0.440
Rumeacet		1								4	4			4			1																			-0.400	
Lathprat				1						1	1																										-0.391
Alloprat	1	1	2		2	3	7	4	7	1	5	5	1	5	4		3		2	1			2	1												-0.378	
Descocosp				1	3					8	5							4	1																		-0.357
Viciorac				2	1					4			1	9		1																					-0.287
Holclana				1						4	6	2	4						1																		-0.219
Festprat				1									2											1													-0.186
Cateprat				1						3	2	1	2		2	1	2	1					1		1												-0.117
Poteanse										1	2	2			1							1															-0.091
Lychflos						1	1				1		2		1								1														-0.059
Trifrepe							1	4	3		4	2	1																								-0.034
Caredist							2															1															-0.015
Junceffu										1	4				1	1										1	1										-0.001
Carevulp										1	1		1	1	3						1	1														0.021	
Poterept												1								1																0.072	
Stelpalu											1	1		1	2	2	1					1	1													0.097	
Poa triv						2					2	3	1		1	2	4							3	1	1										0.099	
Poa palu						1	1				2	6	1	4	6	1	4	1	1	1			1	3												0.117	
Eleounig											1										2															0.124	
Ranurepe						1	1	3	4	4	9	1	4	5	4	2	3	4	1			6	2	9	1											0.131	
RorisyA						1		4		1	1											3		2												0.164	
Agrocani										1		4												5												0.217	
Cirsilanc																																				0.225	
Sympoffi																								1												0.295	
Caltpalu																																				0.322	
Planinte													1	1	1						3														0.361		
Carevesi															4	2	1					7		4											0.363		
Ranufflam						1					1				1							1		3	3											0.383	
Caregrac						2					5				4	1	4				7	3	2	8	2	1									0.399		
Phalarun						1	2		1	1	1	5	1	1	2		2				6	2	1	1	1	8									0.453		
Siumlati															1	2	3						1	2	1										0.476		
Agrostol						6									5	1		4								3	12	8							0.512		
Galipalu						2					2		1	1		1	1		1	3	3	3	2	2								1	1		0.577		
Alopgeni															2				4								5	1	6	3					1.009		
Roriamph																																				1.298	
Glycflui																2												1		4	7				1.753		
Glycmayi																																		1	2	3	3.117
	-1.488									-0.980																											
	-1.441								-0.801																												
	-1.438								-0.656																												
	-1.410								-0.259																												
	-1.303								-0.081																												
	-1.209								-0.456																												
	-0.967								-0.801																												
	-0.861								-0.980																												
	-0.656								-0.980																												
	-0.259								-0.456																												
	-0.222								-0.801																												
	-0.170								-0.980																												
	-0.002								-0.456																												
	0.166								-0.801																												
	0.194								-0.980																												
	0.225								-0.456																												
	0.267								-0.801													</															

The principle:

Calculate the **weighted average of species**
from **site** values (current order of sites)

$$\tilde{x}_j = \frac{\sum_{i=1}^n x_i y_{ij}}{\sum_{i=1}^n y_{ij}}$$

x_j = Weighted average of species j

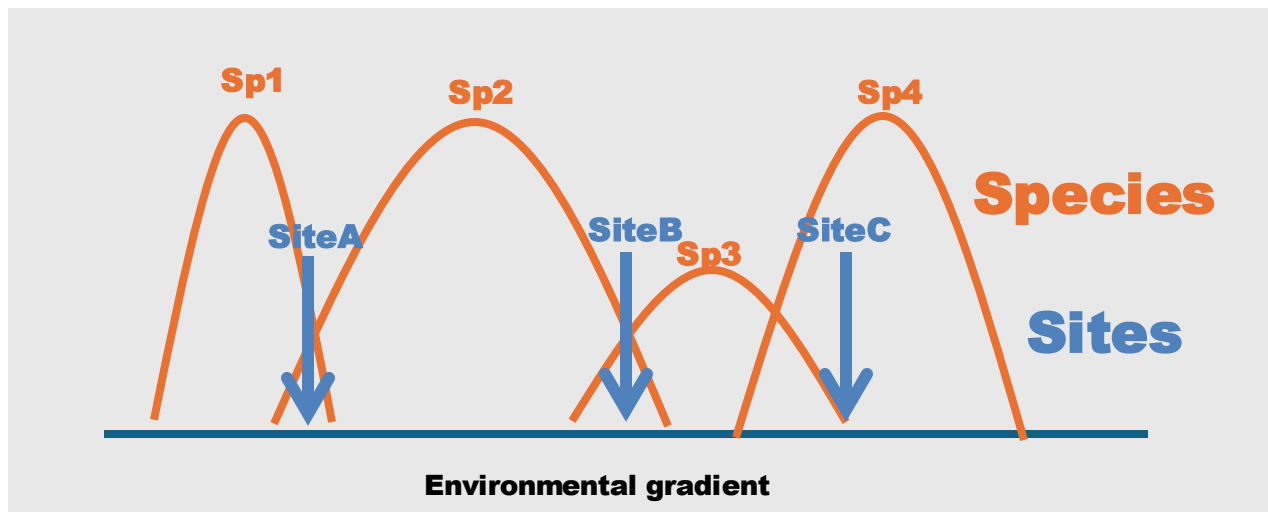
n = Number of sites

y_{ij} = **Abundance** of species j in site i

x_i = **Site score** of site i

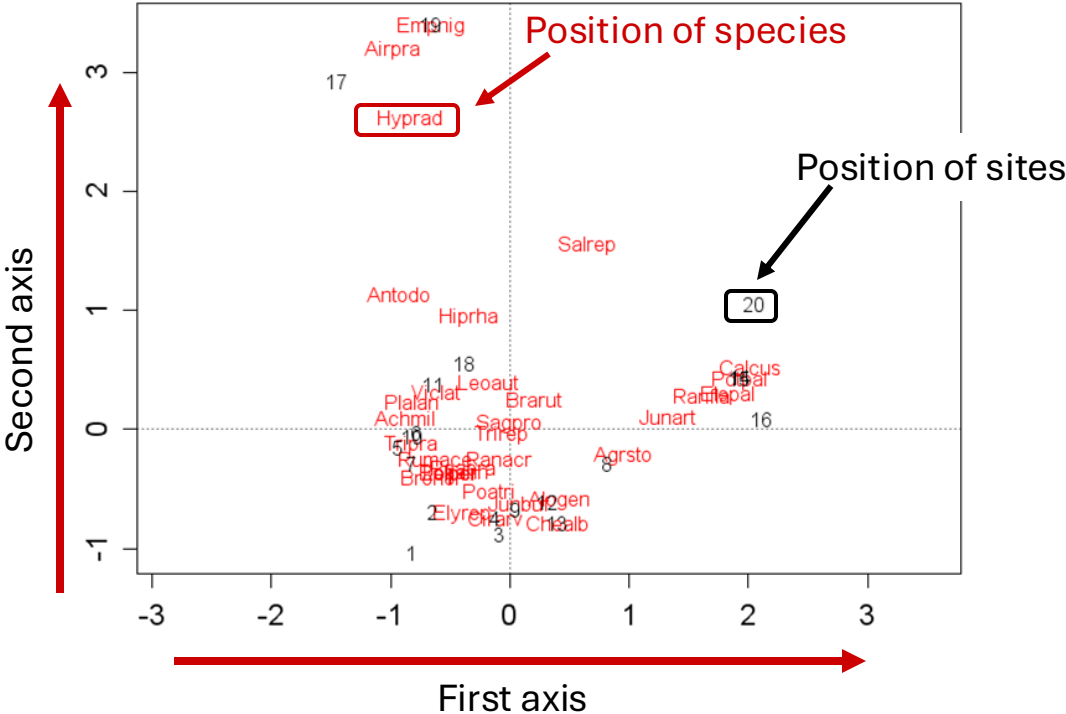
Correspondence Analysis

What is it trying to do?

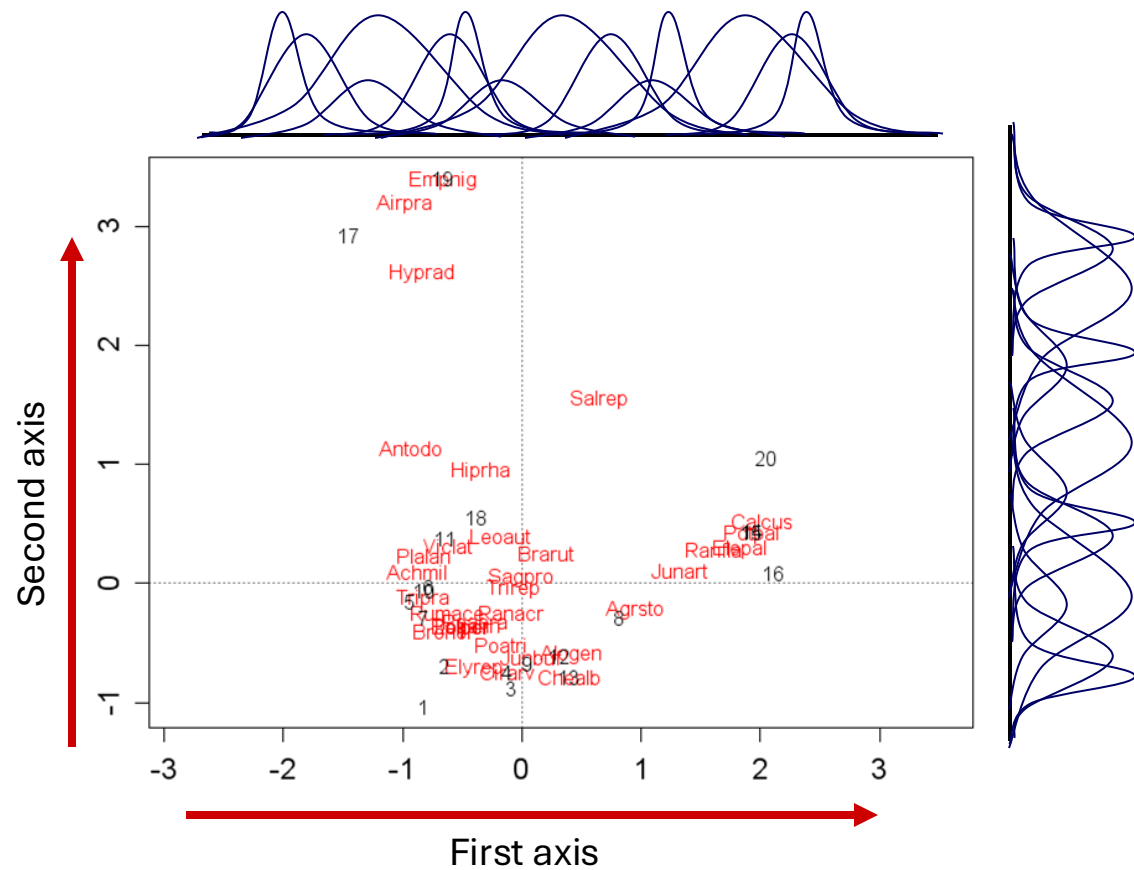


If you knew the species or the sites you could make a guess about the other

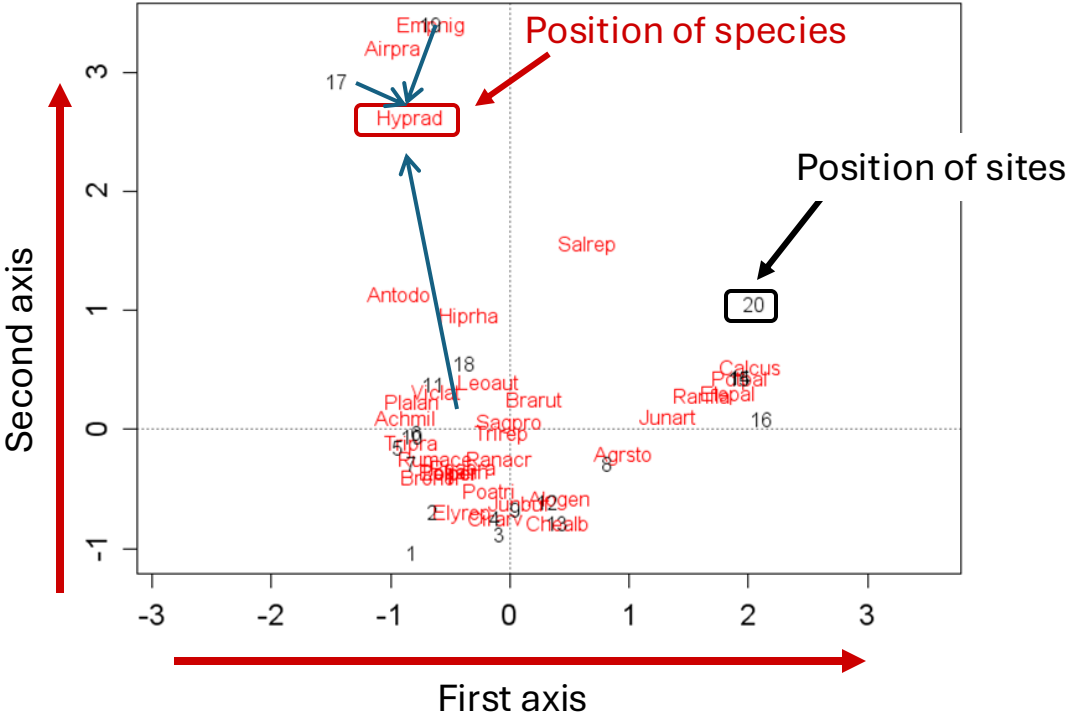
Correspondence Analysis



Correspondence Analysis



Correspondence Analysis



Correspondence Analysis

Each axis of the CA is an **eigenvector**

Each axis has an **eigenvalue**

- Measures the degree of differentiation between species along this axis (range 0-1)
- The larger, the more species can be differentiated along this axis

Total Unconstrained Inertia
2.156 2.156 19

← total inertia: Overall variance of the data

Eigenvalues for unconstrained axes:

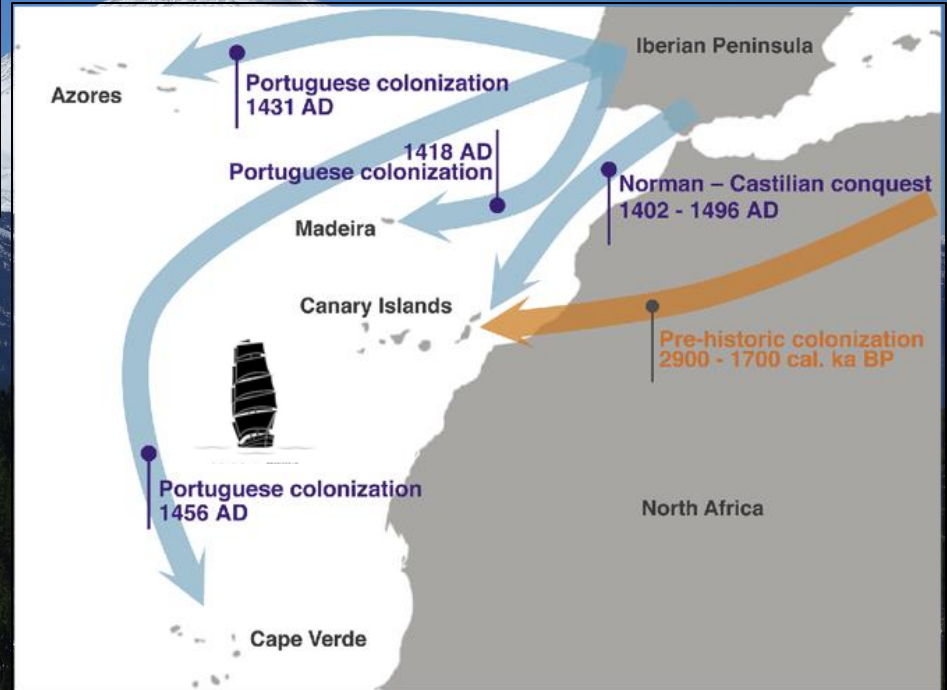
CA1	CA2	CA3	CA4	CA5	CA6
0.52722	0.39537	0.25162	0.20418	0.14553	0.10627

→ Decline in explained variance with increasing ordination axes

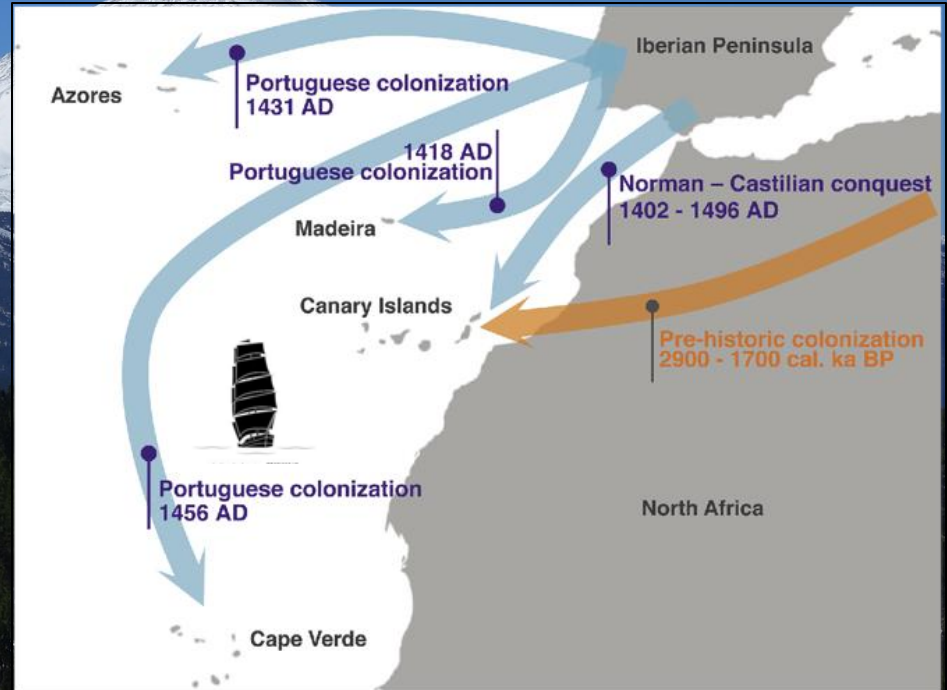


Tenerife, 3718m, Canary Islands, from Christoph Hormann, imagico.de

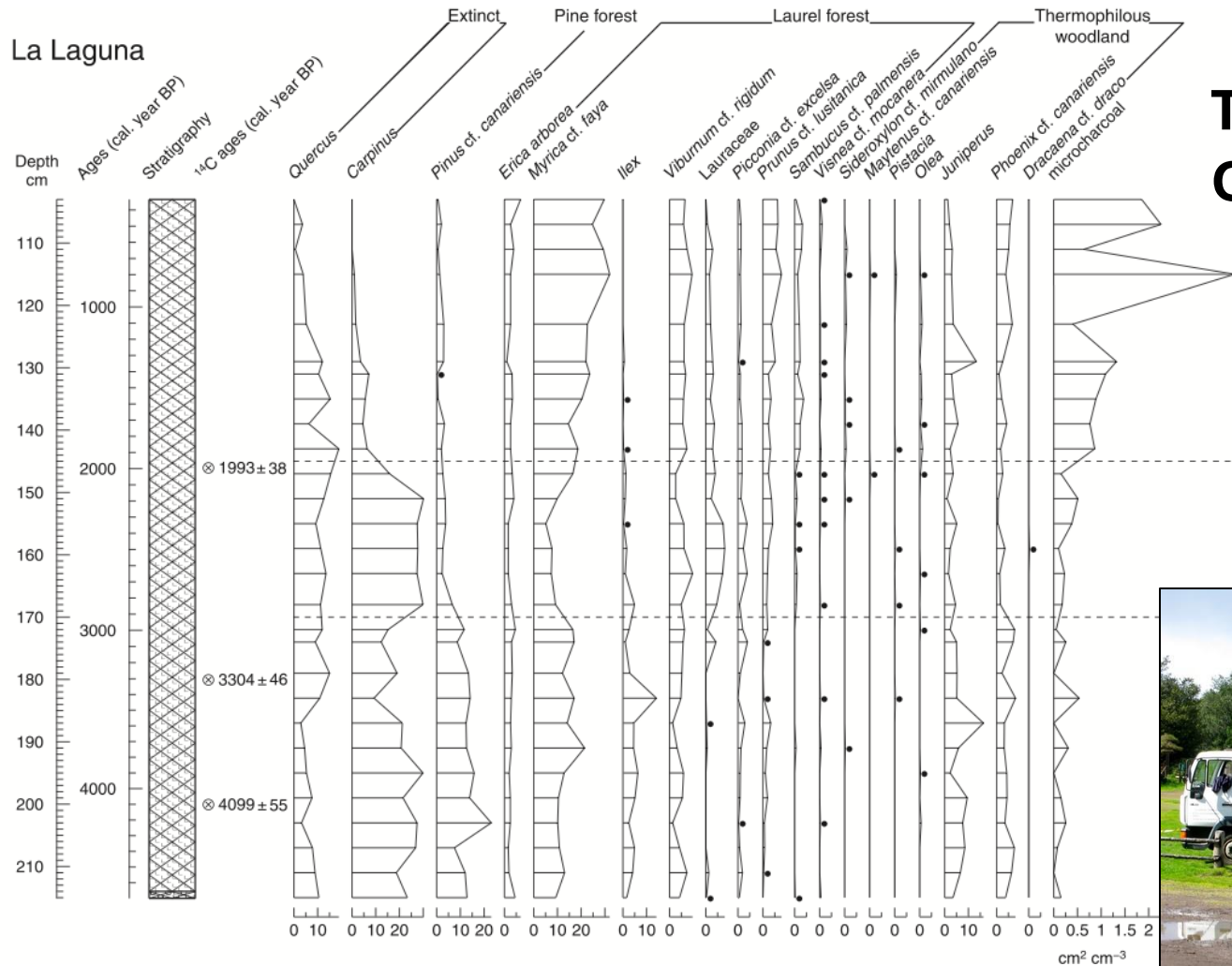
Islands



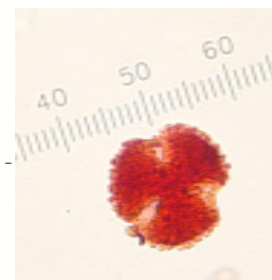
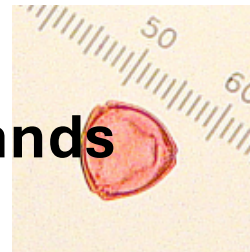
Islands



La Laguna

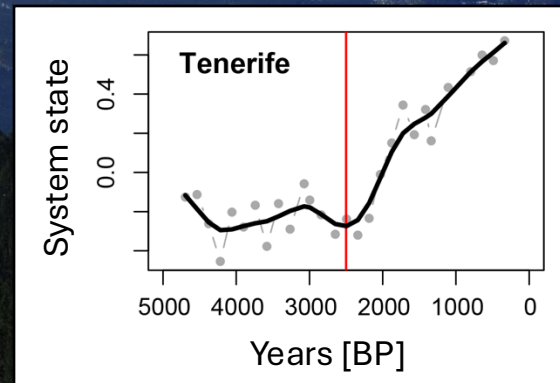


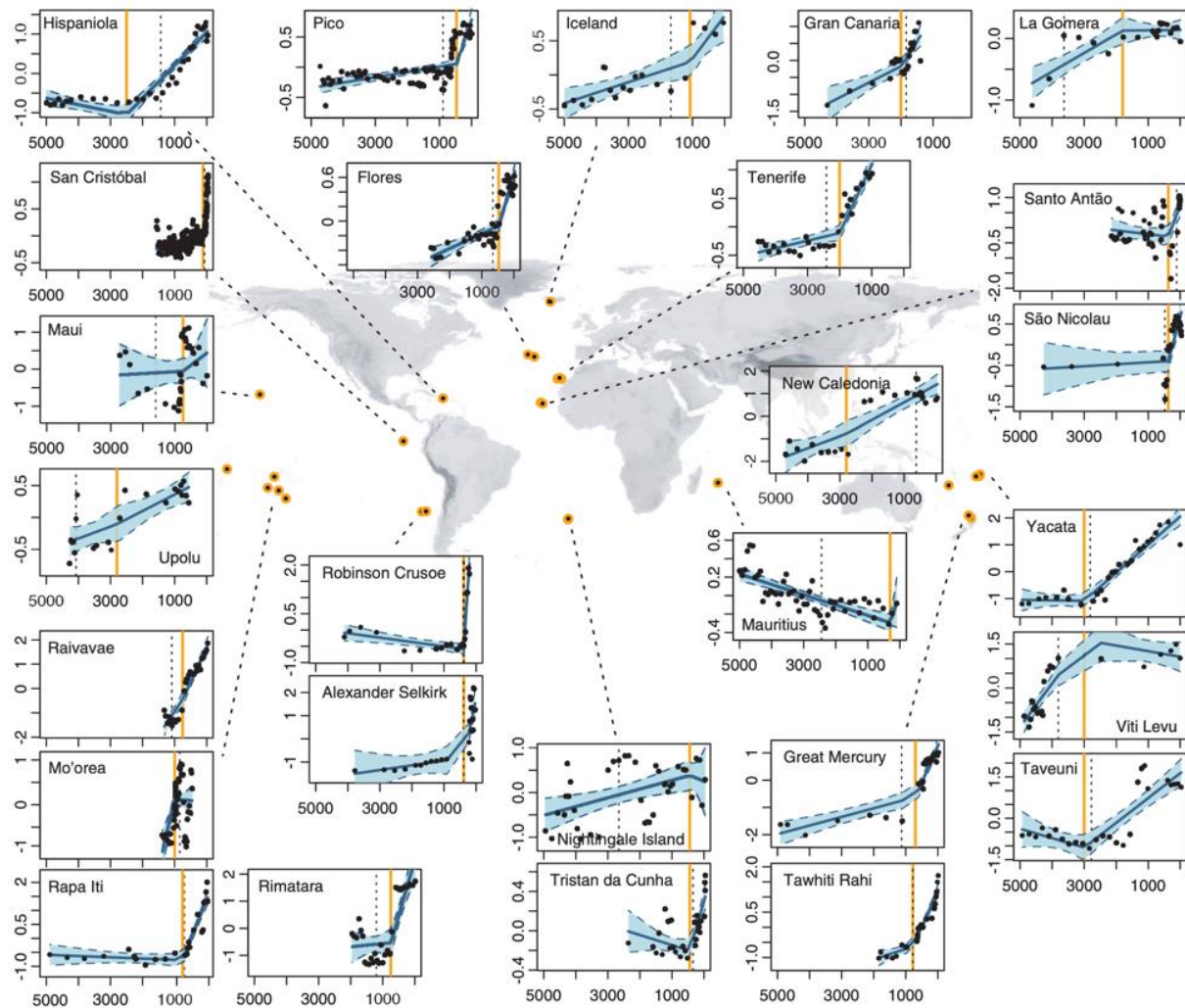
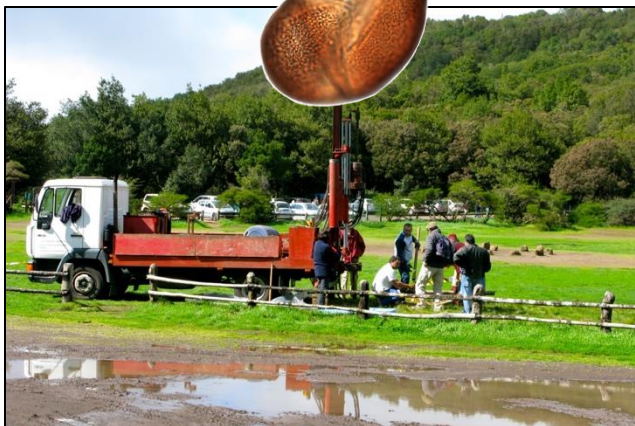
Tenerife, Canary Islands





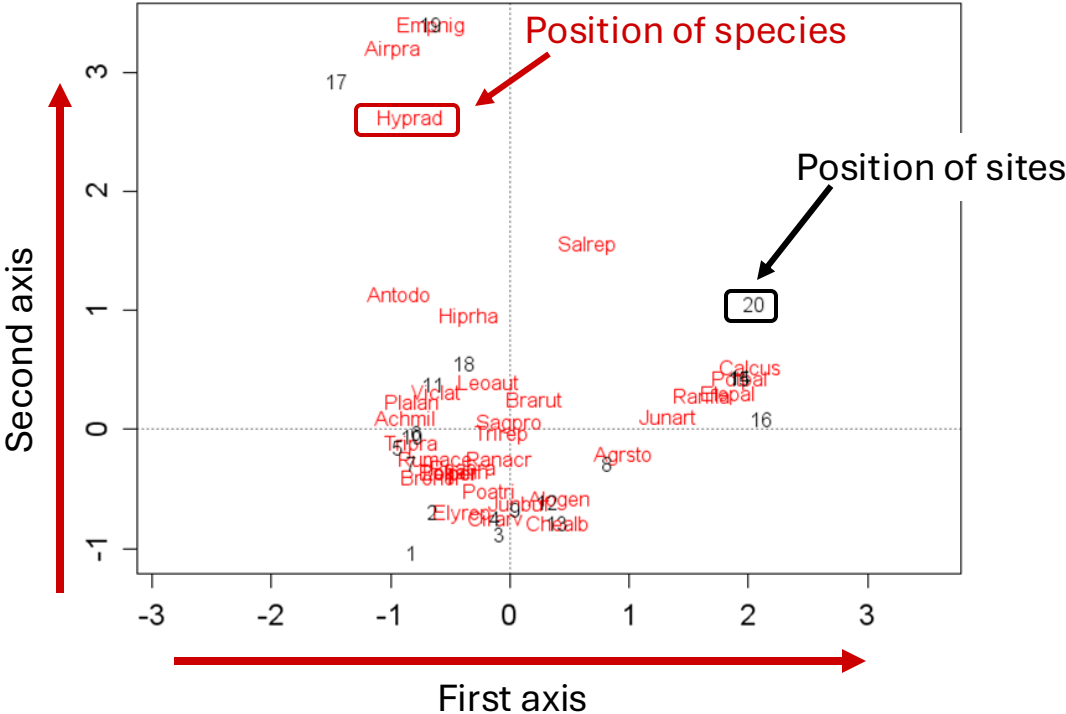
Islands





Nogue et al. (Science, 2021)

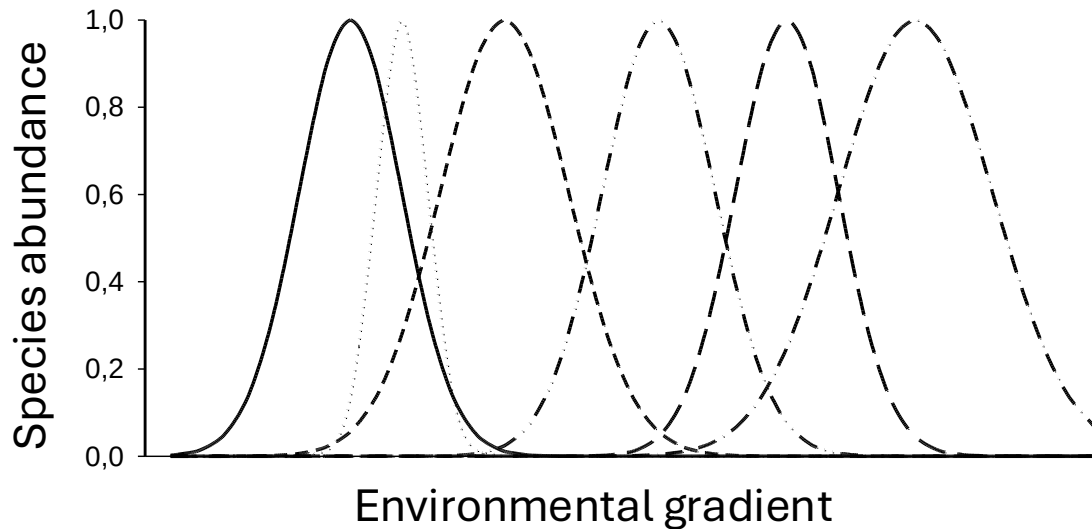
Correspondence Analysis



Ordination

Linear $\leftrightarrow$ unimodal Ordination methods?

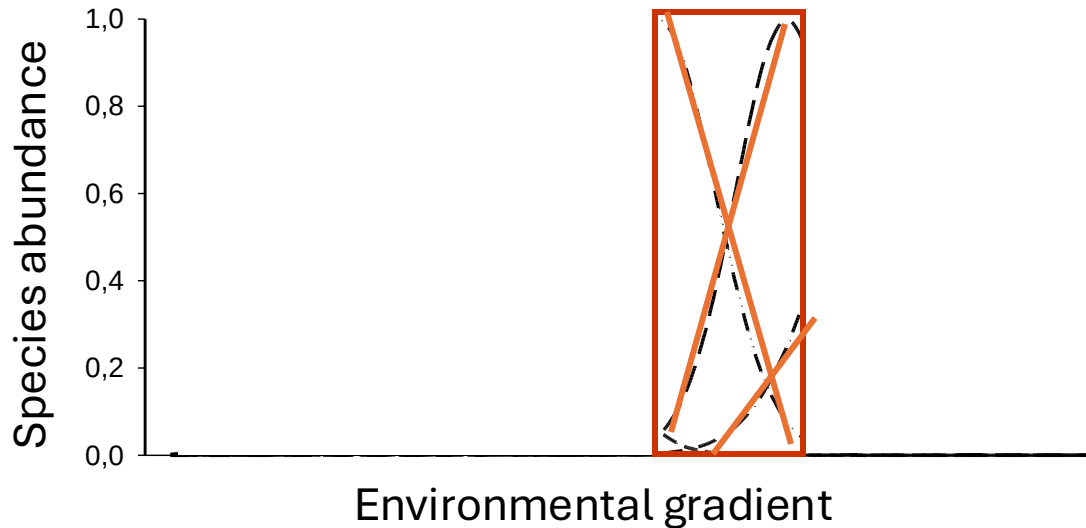
- Depends on lengths of gradient
- Long gradient $\rightarrow$ unimodal „relation“



Ordination

Linear $\leftrightarrow$ unimodal Ordination methods?

- Depends on lengths of gradient
- Long gradient $\rightarrow$ unimodal „relation“



Overview on ordination methods

	Indirect (no environmental data)
Linear relation	PCA (PCoA) Principal Components/ Coordinates Analysis
Unimodel relation	CA (DCA) (Detrended) Correspondence Analysis

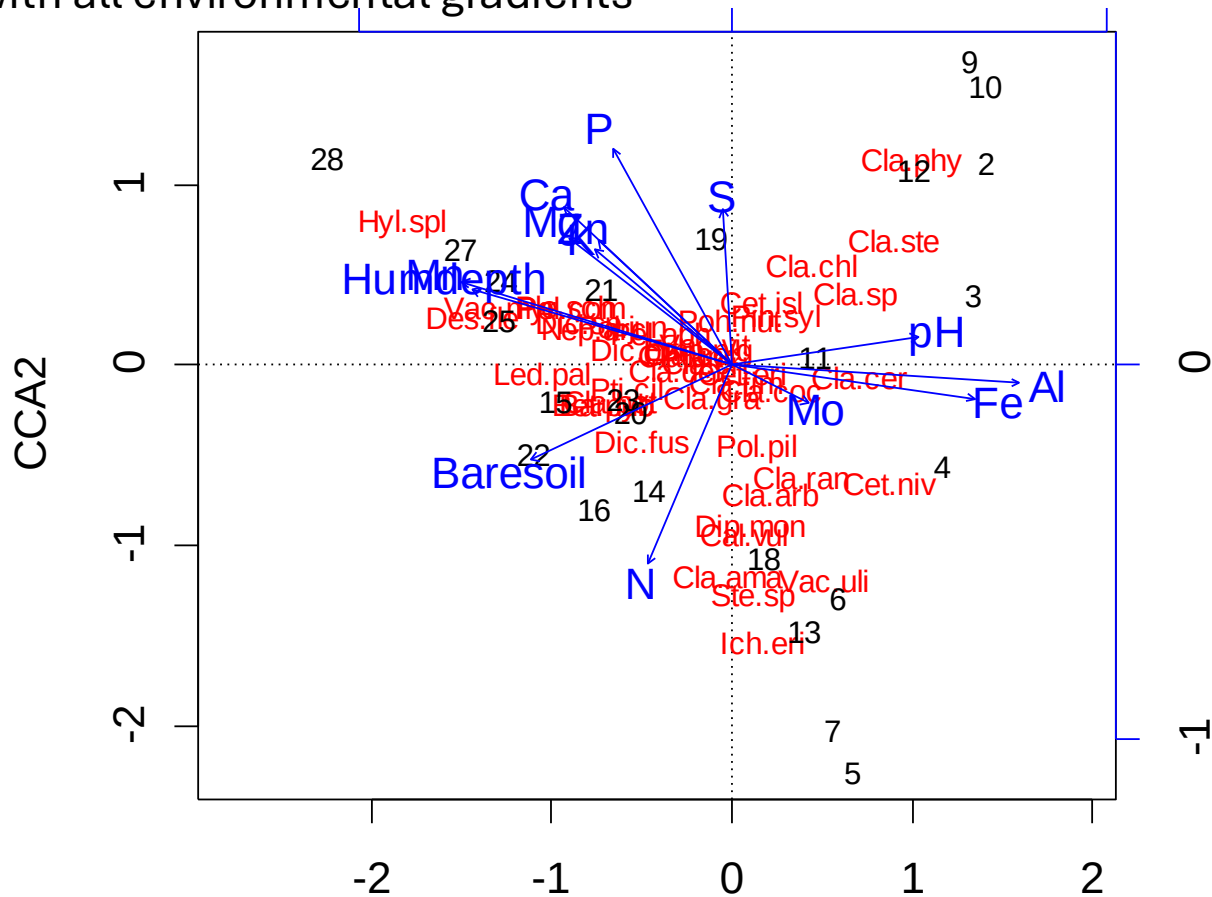
Overview on ordination methods

	Indirect (no environmental data)
Linear relation	PCA (PCoA) Principal Components/ Coordinates Analysis
Unimodel relation	CA (DCA) (Detrended) Correspondence Analysis
Any relation	NMDS (Non-metric multidimensional scaling)

Overview on ordination methods

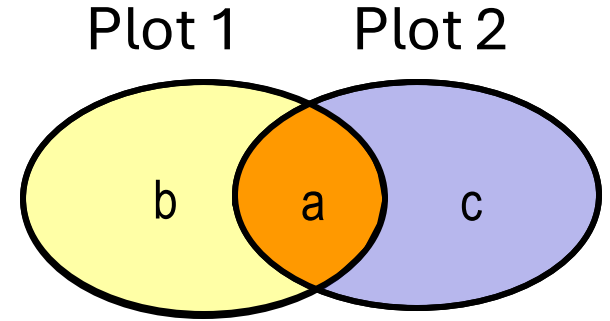
	Indirect (no environmental data)	Direct (with environmental data)
Linear relation	PCA (PCoA) Principal Components/ Coordinates Analysis	RDA Redundancy Analysis
Unimodel relation	CA (DCA) (Detrended) Correspondence Analysis	CCA Canonical CA
Any relation	NMDS (Non-metric multidimensional scaling)	

CCA – with all environmental gradients



Similarity- and dissimilarity indices

		<i>Plot 1</i>	
		+	-
<i>Plot 2</i>	+	a	c
	-	b	d



a: Shared species

b: Plot 1 exclusive species

c: Plot 2 exclusive species

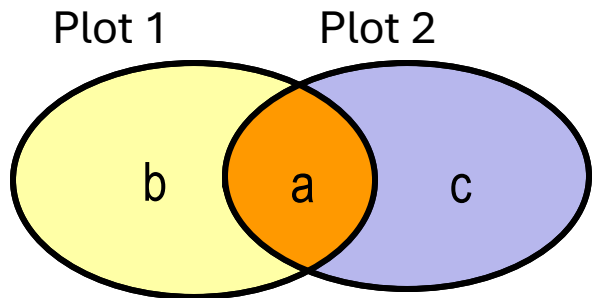
e.g.: Sørensen similarity = $2a / (2a + b + c)$

Correspondence Analysis

Correspondence Analysis implicitly uses Chi-square distance/similarity. Which can be expressed as:

$$sim = \frac{(a+b+c+d)(ad-bc)^2}{(a+b)(a+c)(b+d)(c+d)}$$

Yule & Kendall (1950) in Jurasinski (2012)



a: Shared species

b: Plot 1 exclusive species

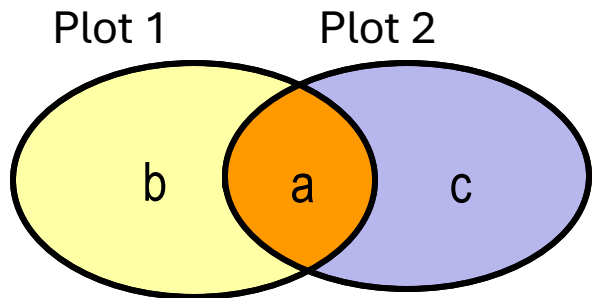
c: Plot 2 exclusive species

d: Species not in the two plots

Correspondence Analysis

Correspondence Analysis implicitly uses Chi-square distance/similarity. Which can be expressed as:

$$\beta_{\text{Sørensen}} = \frac{2a}{2a + b + c}$$



a: Shared species

b: Plot 1 exclusive species

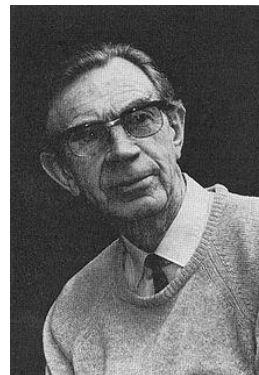
c: Plot 2 exclusive species

d: Species not in the two plots

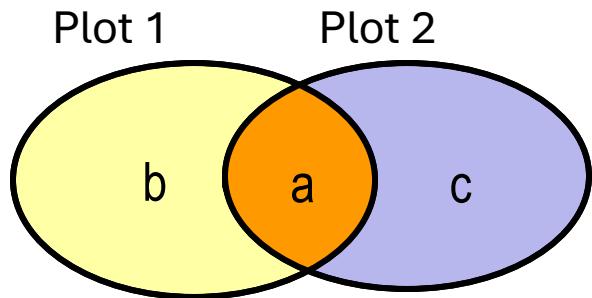
Correspondence Analysis

Correspondence Analysis implicitly uses Chi-square distance/similarity. Which can be expressed as:

$$\beta_{\text{Sørensen}} = \frac{2a}{2a + b + c}$$



Thorvald Sørensen



a: Shared species

b: Plot 1 exclusive species

c: Plot 2 exclusive species

d: Species not in the two plots

(Dis)Similarity Matrices

	v_1	v_2	v_3	v_4
o_1	$v_{1,1}$	$v_{1,2}$	$v_{1,3}$	$v_{1,4}$
o_2	$v_{2,1}$	$v_{2,2}$	$v_{2,3}$	$v_{2,4}$
o_3	$v_{3,1}$	$v_{3,2}$	$v_{3,3}$	$v_{3,4}$
o_4	$v_{4,1}$	$v_{4,2}$	$v_{4,3}$	$v_{4,4}$
o_5	$v_{5,1}$	$v_{5,2}$	$v_{5,3}$	$v_{5,4}$
o_6	$v_{6,1}$	$v_{6,2}$	$v_{6,3}$	$v_{6,4}$

Raw data



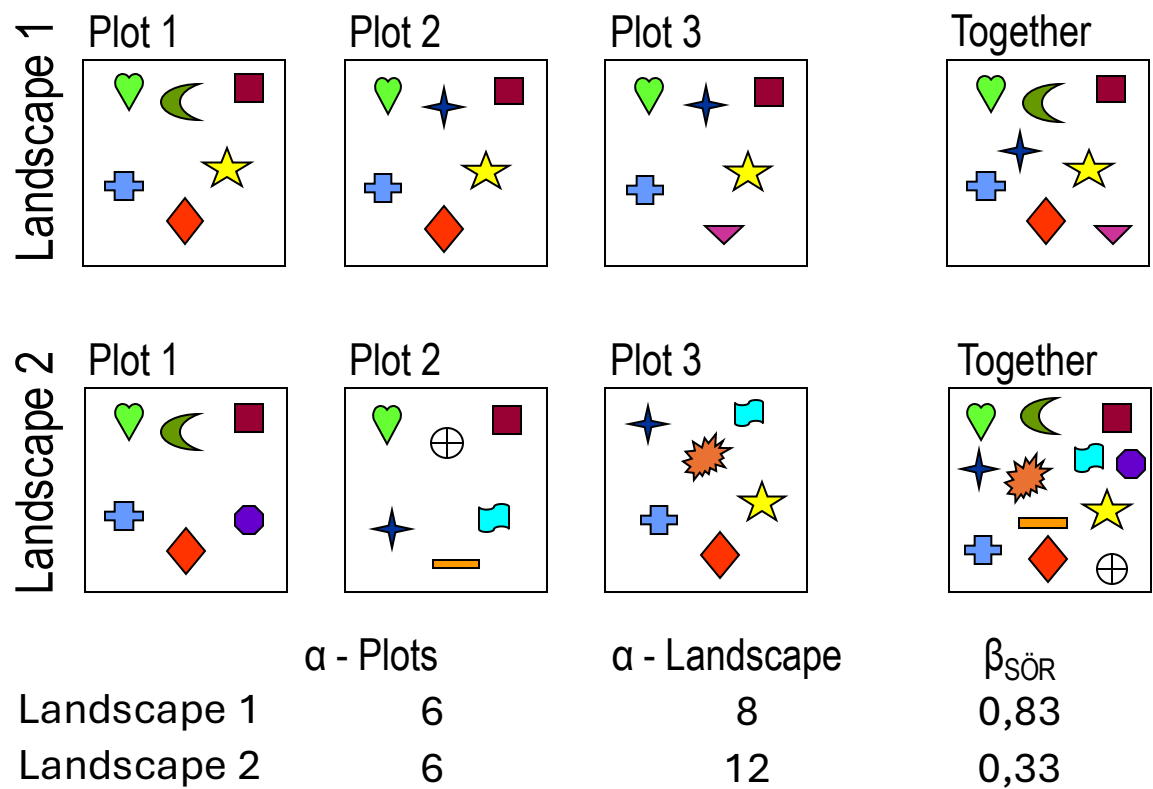
dissimilarity matrix

o_1	0					
o_2	$d_{2,1}$	0				
o_3	$d_{3,1}$	$d_{3,2}$	0			
o_4	$d_{4,1}$	$d_{4,2}$	$d_{4,3}$	0		
o_5	$d_{5,1}$	$d_{5,2}$	$d_{5,3}$	$d_{5,4}$	0	
o_6	$d_{6,1}$	$d_{6,2}$	$d_{6,3}$	$d_{6,4}$	$d_{6,5}$	0

	o_1	o_2	o_3	o_4	o_5	o_6
o_1	1					
o_2	$s_{2,1}$	1				
o_3	$s_{3,1}$	$s_{3,2}$	1			
o_4	$s_{4,1}$	$s_{4,2}$	$s_{4,3}$	1		
o_5	$s_{5,1}$	$s_{5,2}$	$s_{5,3}$	$s_{5,4}$	1	
o_6	$s_{6,1}$	$s_{6,2}$	$s_{6,3}$	$s_{6,4}$	$s_{6,5}$	1

similarity matrix

Why do we need these measures?



Diversity of indices

$$\beta_{\text{Sørensen}} = \frac{2a}{2a + b + c}$$

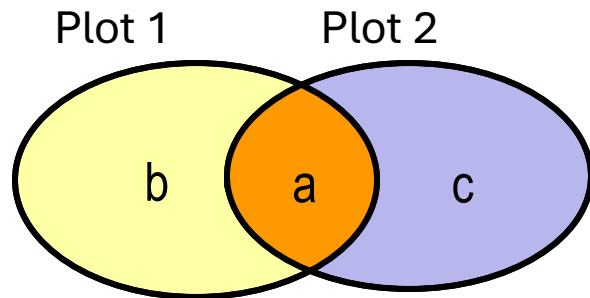
$$\beta_{\text{Williams}} = \frac{\min(b, c)}{a + b + c}$$

$$\beta_{\text{Jaccard}} = \frac{a}{a + b + c}$$

$$\beta_{\text{Lande}} = \frac{b + c}{2}$$

$$\beta_{\text{Wilson \& Shmida}} = \frac{b + c}{2a + b + c}$$

Diversity of indices



soerensen

$$sim = \frac{2a}{2a+b+c}$$

jaccard

$$sim = \frac{a}{a+b+c}$$

ochiai

$$sim = \frac{a}{\sqrt{(a+b)(a+c)}}$$

mountford

$$sim = \frac{2a}{(a+b+c)+2bc}$$

whittaker

$$sim = \frac{a+b+c}{\frac{2a+b+c}{2}} - 1$$

lande

$$sim = \frac{b+c}{2}$$

wilsonshmida

$$sim = \frac{b+c}{2a+b+c}$$

cocogaston

$$sim = \frac{b+c}{a+b+c}$$

magurran

$$sim = (2a+b+c)(1 - \frac{a}{a+b+c})$$

harrison

$$sim = \frac{\min(b,c)}{\max(b,c)+a}$$

cody

$$sim = 1 - \frac{a(2a+b+c)}{2(a+b)(a+c)}$$

williams

$$sim = \frac{\min(b,c)}{a+b+c}$$

williams2

$$\frac{(bc)+1}{(a+b+c)^2 - (a+b+c)}$$

harte

$$1 - \frac{2a}{2a+b+c}$$

simpson

$$\frac{\min(b,c)}{\min(b,c)+a}$$

lennon

$$\frac{2|b-c|}{2a+b+c}$$

weiher

$$sim = b + c$$

ruggiero

$$sim = \frac{a}{a+c}$$

lennon2

$$sim = 1 - \left[\frac{\log\left(\frac{2a+b+c}{a+b+c}\right)}{\log 2} \right]$$

routledge

$$sim = \frac{(a+b+c)^2}{(a+b+c)^2 - 2bc}$$

rout2ledge

toolong, seebelow

rout3ledge

$$sim = e^{rout2ledge} - 1$$

sokal1

$$sim = \frac{a}{a+2(b+c)}$$

dice

$$sim = \frac{a}{\min((b+a), (c+a))}$$

kulczlinsky

$$sim = \frac{a}{b+c}$$

Soerensen (1948)

Jaccard (1912)

Ochiai (1957), Shi (1993)

Mountford (1962), Shi (1993)

Whittaker (1960), Magurran (1988)

Lande (1996)

Wilson & Shmida (1984)

Colwell & Coddington (1948), Gaston et al. (2001)

Magurran (1988)

Harrison et al. (1992), Koleff et al. (2003)

Cody (1993)

Williams (1996), Koleff et al. (2003)

Williams (1996), Koleff et al. (2003)

Harte & Kinzig (1997), Koleff et al. (2003)

Simpson (1949), Koleff et al. (2003)

Lennon et al. (2001), Koleff et al. (2003)

Weiher & Boylen (1994)

Ruggiero et al. (1998), Koleff et al. (2003)

Lennon et al. (2001), Koleff et al. (2003)

Lennon et al. (2001), Koleff et al. (2003)

Routledge (1977), Magurran (1988)

Routledge (1977), Wilson & Shmida (1984)

Routledge (1977)

Sokal & Sneath (1963)

Association index of Dice (1945), Wolda (1981)

Oosting (1956), Southwood (1978)

Diversity of indices

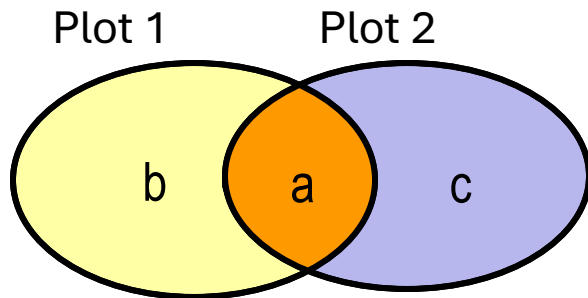
kulcz2insky	$sim = \frac{\frac{3}{2}(2a+b+c)}{(a+b)(a+c)}$
mcconnagh	$sim = \frac{a^2 - bc}{(a+b)(a+c)}$
simpson2	$sim = \frac{a}{a+b}$
legendre2	$sim = \frac{3a}{3a+b+c}$
fager	$sim = \frac{1}{\sqrt{n_1 n_2} - 2\sqrt{n_2}}$
maarel	$sim = \frac{2a - (b+c)}{2a+b+c}$
lamont	$sim = \frac{a}{2a+b+c}$
johnson	$sim = \frac{a^2}{2b}$
sorgenfrei	$sim = \frac{a^2}{(a+b)(a+c)}$
johnson2	$sim = \frac{a}{a+b} + \frac{a}{a+c}$

Oosting (1956), Southwood (1978)
 Hubalek (1982)
 Simpson (1960), Shi (1993)
 Legendre & Legendre (1998)
 Fager (1957), Shi (1993)
 van der Maarel (1969)
 Lamont and Grant (1979)
 Johnson (1971)
 Sorgenfrei (1959)
 Johnson (1967)

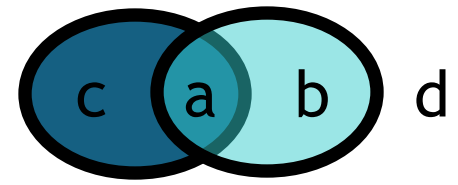
Computable symmetric indices (including unshared species):

manhattan	$sim = \frac{b+c}{a+b+c+d}$
simplematching	$sim = \frac{a+d}{a+b+c+d}$
margaleff	$sim = \frac{a(a+b+c+d)}{(a+b)(a+c)}$
pearson	$sim = \frac{ad-bc}{\sqrt{(a+b)(a+c)(d+b)(d+c)}}$
roger	$sim = \frac{a+d}{a+2(b+c)+d}$
baroni	$sim = \frac{\sqrt{ad+c}}{\sqrt{ad+a+b+c}}$
dennis	$sim = \frac{ad-bc}{\sqrt{(a+b+c+d)(a+b)(a+c)}}$
fossum	$sim = \frac{(a+b)(a+c)}{(a+b+c+d)(-\frac{3}{2})^2}$
gower	$sim = \frac{a - (b+c)+d}{a+b+c+d}$
legendre	$sim = \frac{ad}{a+b+c+d}$
sokal2	$sim = \frac{ad}{\sqrt{(a+b)(a+c)(d+b)(d+c)}}$
sokal3	$sim = \frac{2ad+2d}{(a+d)+(a+b+c+d)}$
sokal4	$sim = \frac{a+d}{b+c}$
stiles	$sim = \frac{(a+b+c+d)(ad-bc - \frac{a+b+c+d}{2})^2}{(a+b)(a+c)(b+d)(c+d)}$
yule	$sim = \frac{ad-bc}{ad+bc}$
michael	$sim = \frac{k(ad-bc)}{(a+b)^2 + (b+c)^2}$
hamann	$sim = \frac{(aN-2n_2)}{(Nn_1-2n_2)}$
forbes	$sim = \frac{2(ad-bc)^2}{(a+b)(a+c)(b+d)(c+d)}$
chisquare	$sim = \frac{(ad-bc)}{(a+c)(b+d)}$
peirce	$sim = \frac{a - (a+b)(a+c)}{(a+b)(a+c)(b+d)(c+d)}$
eyraud	$sim = \frac{a - (a+b)(a+c)}{(a+b)(a+c)(b+d)(c+d)}$

Mean Manhattan, Legendre & Legendre (1998)
 Sokal & Michener 1958
 Clifford & Stevenson (1975)
 Phi of Pearson, Gower & Legendre (1986)
 Rogers & Tanimoto (1960), Gower & Legendre (1986)
 Baroni-Urbani & Buser (1976), Wolda (1981)
 Holliday et al. (2002)
 Holliday et al. (2002)
 Gower & Legendre (1986)
 Gower & Legendre (1986)
 Sokal & Sneath (1963)
 Sokal & Sneath (1963)
 Sokal & Sneath (1963)
 Stiles (1946)
 Yule & Kendall (1973)
 Michael (1920), Shi (1993)
 Hamann (1961)
 Forbes (1925), Shi (1993)
 Yule & Kendall (1950)
 Peirce (1984) in Shi (1993)
 Eyraud (1936) in Shi (1993)



Diversity of indices



$$\beta_{ij} = (\varepsilon a + \delta d) (\varepsilon a + \delta d + \lambda(b + c))^{-1}$$

β_{ij} = Similarity Value for Data Sets i, j

a = Species common in i and j

b = Species only in i

c = Species only in j

d = Species neither in i nor in j but in the area

$\varepsilon, \delta, \lambda$ = coefficients

For $\varepsilon = 1$; $\delta = 0$ and $\lambda = 1$	„Jaccard“-Index,
For $\varepsilon = 2$; $\delta = 0$ and $\lambda = 1$	„Sørensen“-Index,
For $\varepsilon = 1$; $\delta = 1$ and $\lambda = 1$	„Simple Matching“-Index.

Sørensen's Coefficient is binary

Object 1
1 0

Object 2	1	a	b
	0	c	d

$$S = \frac{2a}{2a + b + c}$$

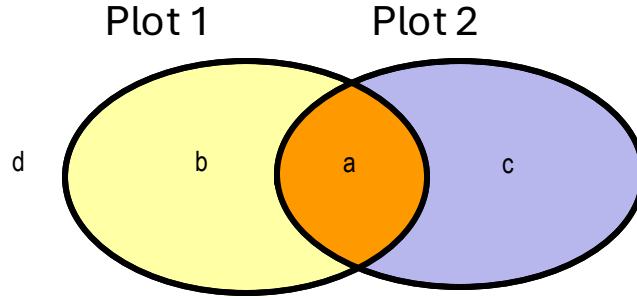
Bray-Curtis (=Steinhaus' Coefficient)

Bray-Curtis is Sørensen coefficient but including abundance information:

$$BC_{ij} = \frac{2C_{ij}}{S_i + S_j}$$

C = sum of the lesser abundance values for shared species
S = total abundance of species counted at both sites

Similarity- and dissimilarity indices



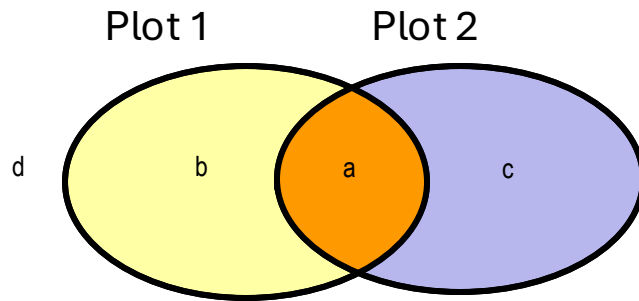
a: Shared species

b: Plot 1 exclusive species

c: Plot 2 exclusive species

d: Further species

Similarity- and dissimilarity indices



a: Shared species

b: Plot 1 exclusive species

c: Plot 2 exclusive species

d: Further species

e.g.: Sørensen similarity = $2a / (2a + b + c)$

	v_1	v_2	v_3	v_4
o_1	$V_{1,1}$	$V_{1,2}$	$V_{1,3}$	$V_{1,4}$
o_2	$V_{2,1}$	$V_{2,2}$	$V_{2,3}$	$V_{2,4}$
o_3	$V_{3,1}$	$V_{3,2}$	$V_{3,3}$	$V_{3,4}$
o_4	$V_{4,1}$	$V_{4,2}$	$V_{4,3}$	$V_{4,4}$
o_5	$V_{5,1}$	$V_{5,2}$	$V_{5,3}$	$V_{5,4}$
o_6	$V_{6,1}$	$V_{6,2}$	$V_{6,3}$	$V_{6,4}$

6

Raw data



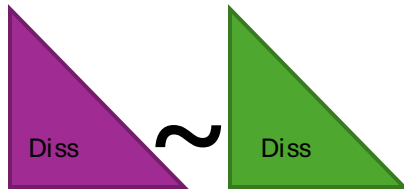
similarity
matrix

	o_1	o_2	o_3	o_4	o_5	o_6
o_1	1					
o_2	$s_{2,1}$	1				
o_3	$s_{3,1}$	$s_{3,2}$	1			
o_4	$s_{4,1}$	$s_{4,2}$	$s_{4,3}$	1		
o_5	$s_{5,1}$	$s_{5,2}$	$s_{5,3}$	$s_{5,4}$	1	
o_6	$s_{6,1}$	$s_{6,2}$	$s_{6,3}$	$s_{6,4}$	$s_{6,5}$	1

6

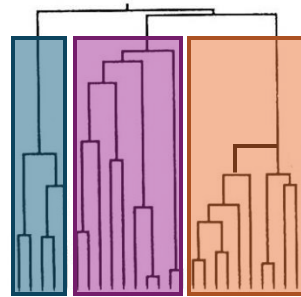
	o_1	o	o_3	o_4	o_5	o_6
o_1	0	2				
o_2	$d_{2,1}$	0				
o_3	$d_{3,1}$	$d_{3,2}$	0			
o_4	$d_{4,1}$	$d_{4,2}$	$d_{4,3}$	0		
o_5	$d_{5,1}$	$d_{5,2}$	$d_{5,3}$	$d_{5,4}$	0	
o_6	$d_{6,1}$	$d_{6,2}$	$d_{6,3}$	$d_{6,4}$	$d_{6,5}$	0

Dissimilarity matrix



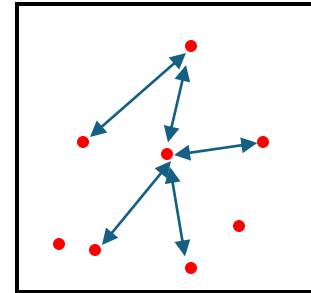
Relate to other
dissimilarity matrix

or



Classification

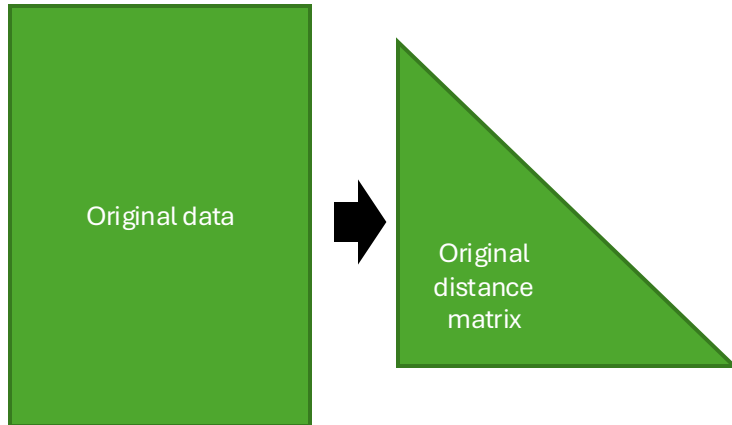
or



Dimension reduction
(e.g. NMDS)

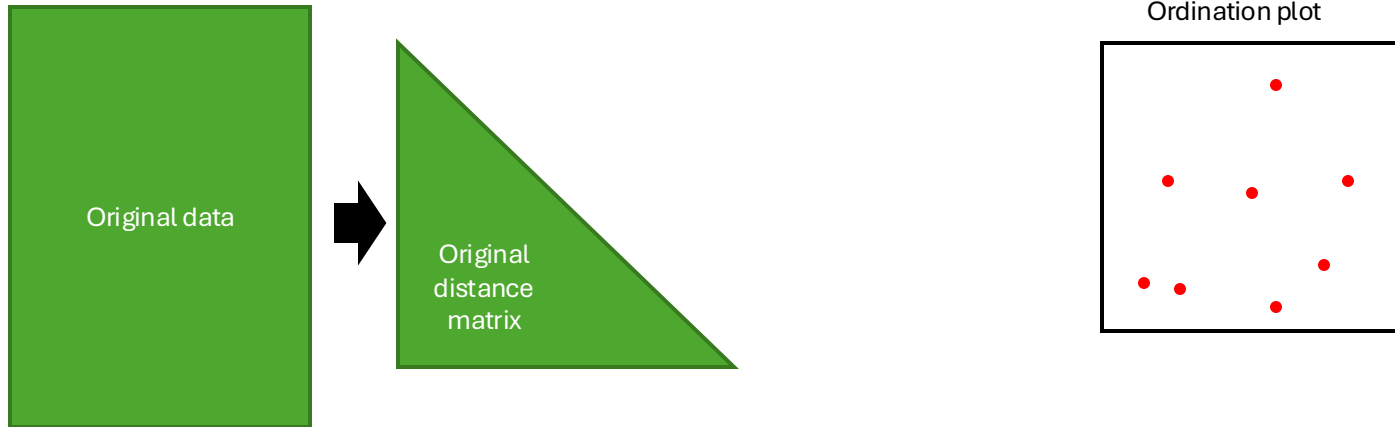
(N)MDS - The goal

- Visualization!
- Create 2d or 3d plot that tries to show the original distances as well as possible



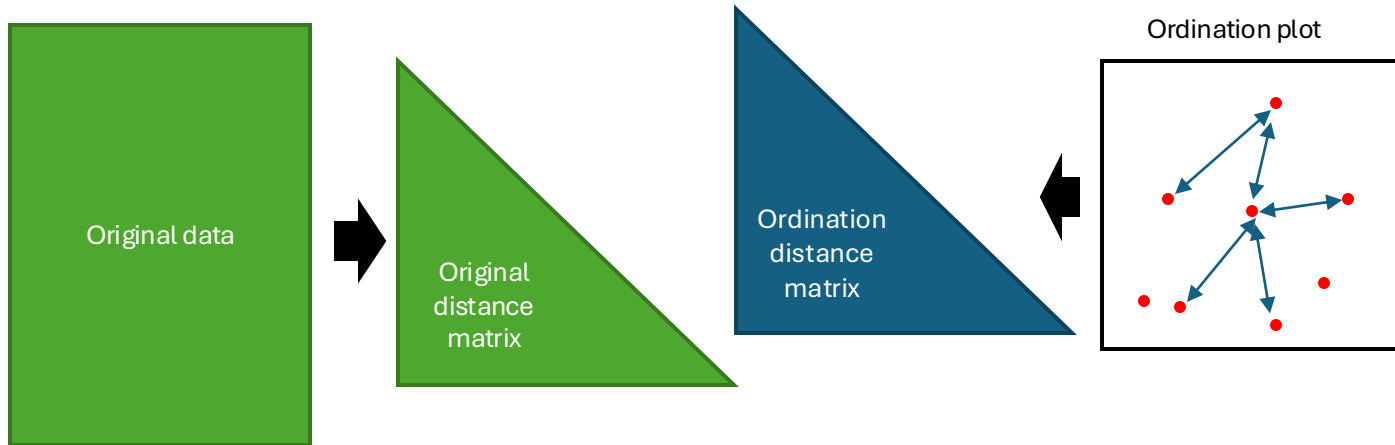
(N)MDS - The goal

- Visualization!
- Create 2d or 3d plot that tries to show the original distances as well as possible



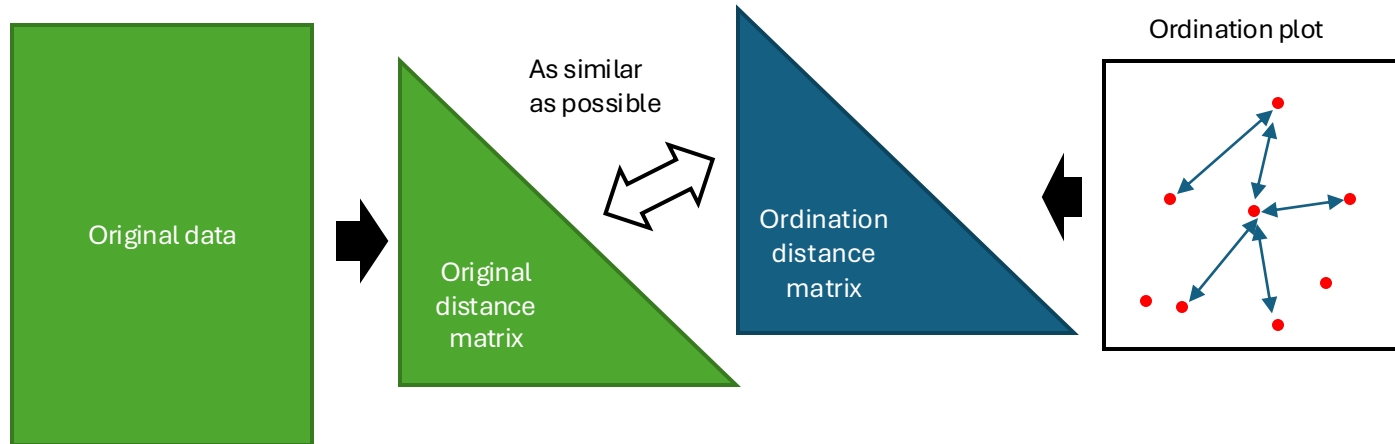
(N)MDS - The goal

- Visualization!
- Create 2d or 3d plot that tries to show the original distances as well as possible

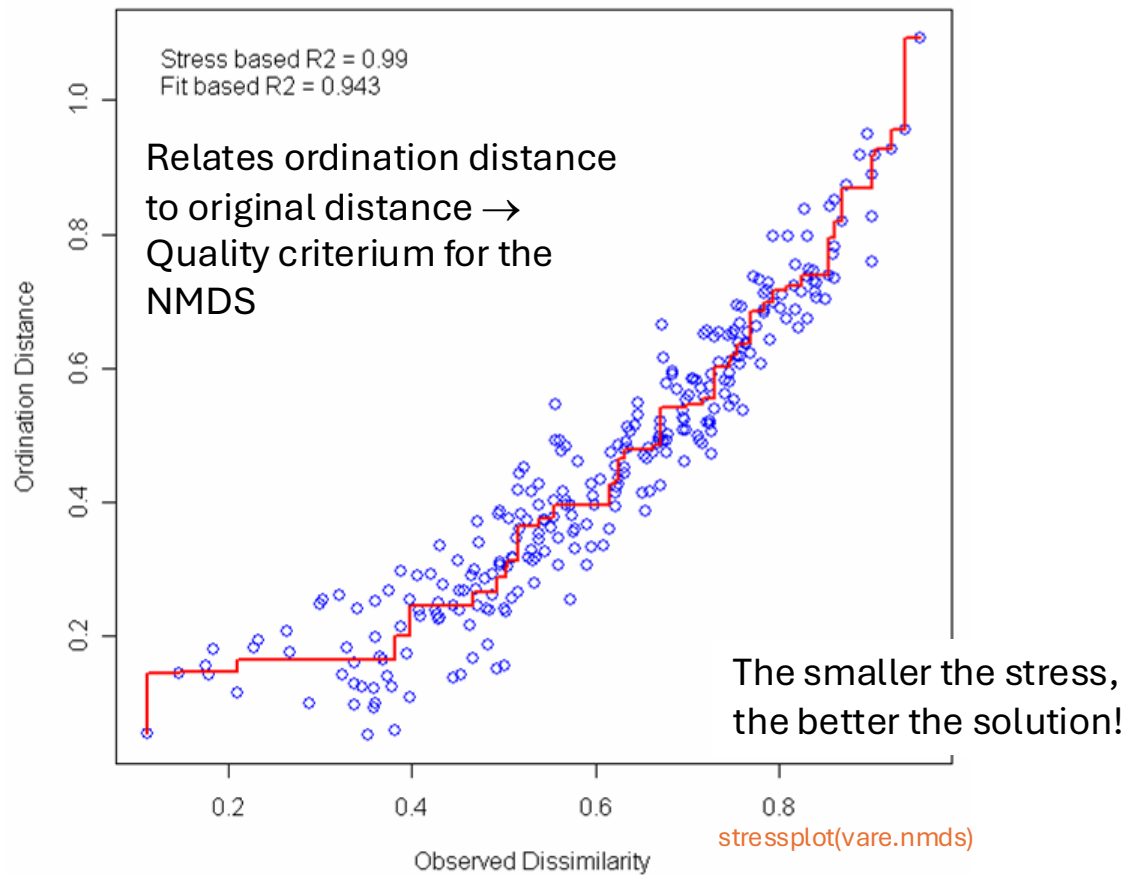


(N)MDS - The goal

- Visualization!
- Create 2d or 3d plot that tries to show the original distances as well as possible



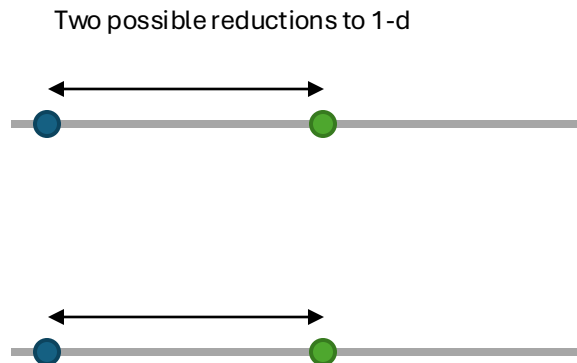
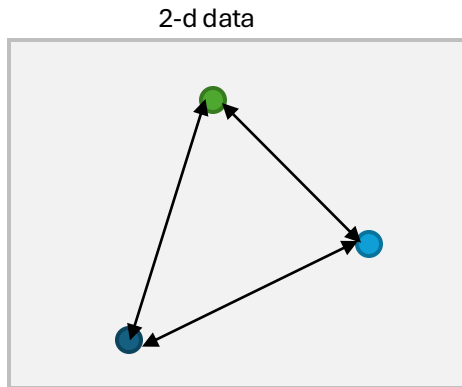
NMDS – stressplot



Ordination

Distortion:

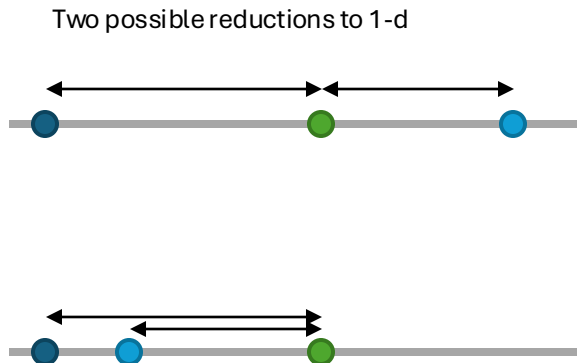
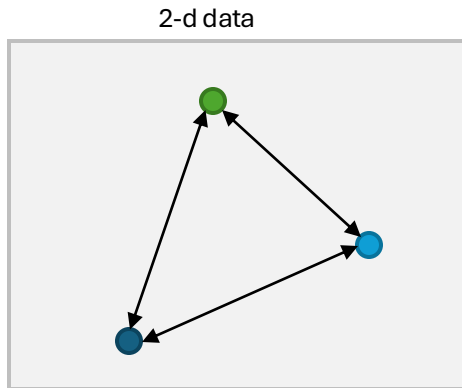
- Reducing high dimensional data to low dimensional data causes loss of information:



Ordination

Distortion:

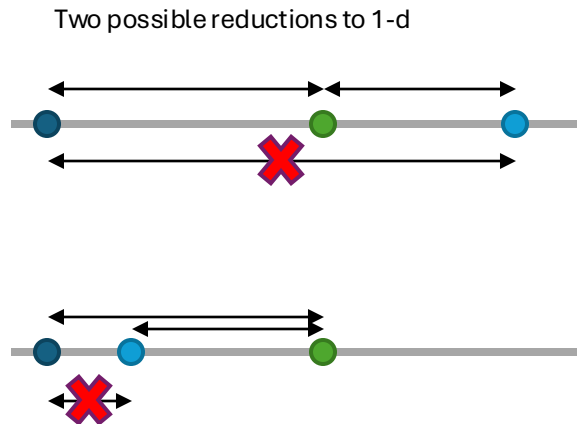
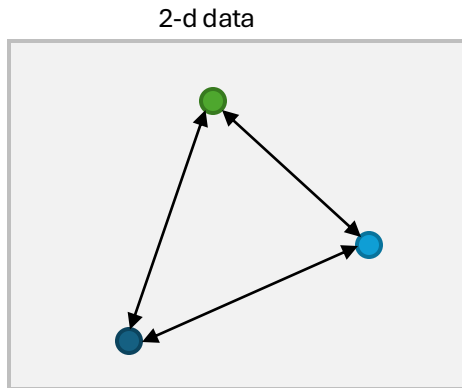
- Reducing high dimensional data to low dimensional data causes loss of information:



Ordination

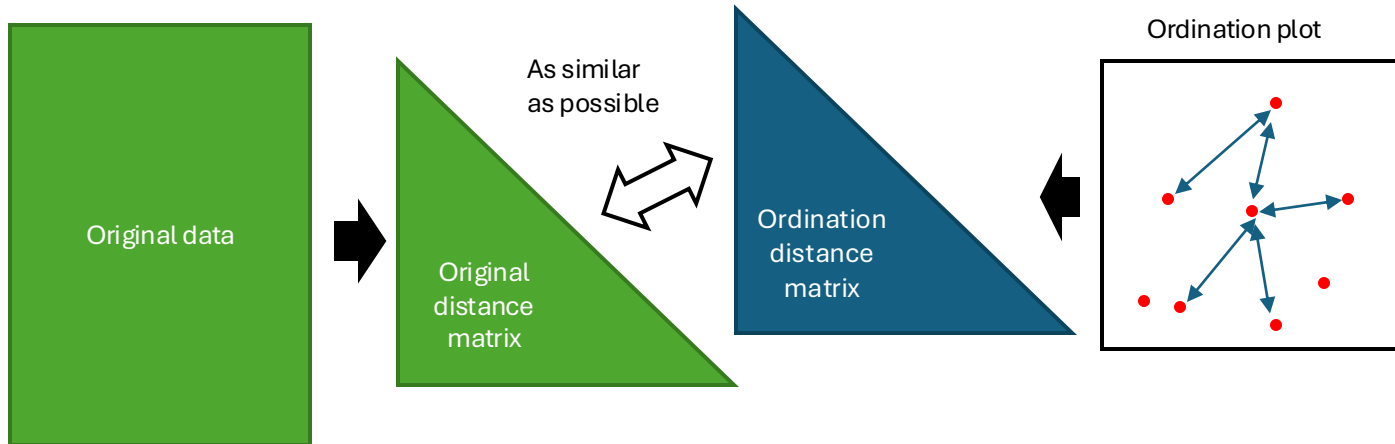
Distortion:

- Reducing high dimensional data to low dimensional data causes loss of information:



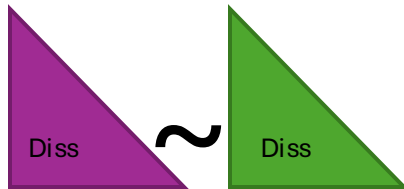
(N)MDS

- Visualization!
- Create 2d or 3d plot that tries to show the original distances as well as possible



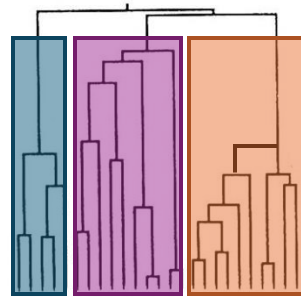
	o_1	o	o_3	o_4	o_5	o_6
o_1	0	2				
o_2	$d_{2,1}$	0				
o_3	$d_{3,1}$	$d_{3,2}$	0			
o_4	$d_{4,1}$	$d_{4,2}$	$d_{4,3}$	0		
o_5	$d_{5,1}$	$d_{5,2}$	$d_{5,3}$	$d_{5,4}$	0	
o_6	$d_{6,1}$	$d_{6,2}$	$d_{6,3}$	$d_{6,4}$	$d_{6,5}$	0

Dissimilarity matrix



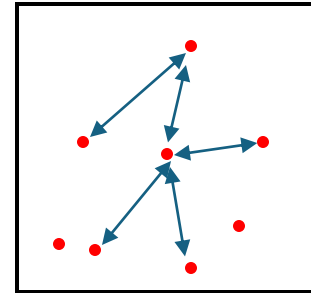
Relate to other
dissimilarity matrix

or



Classification

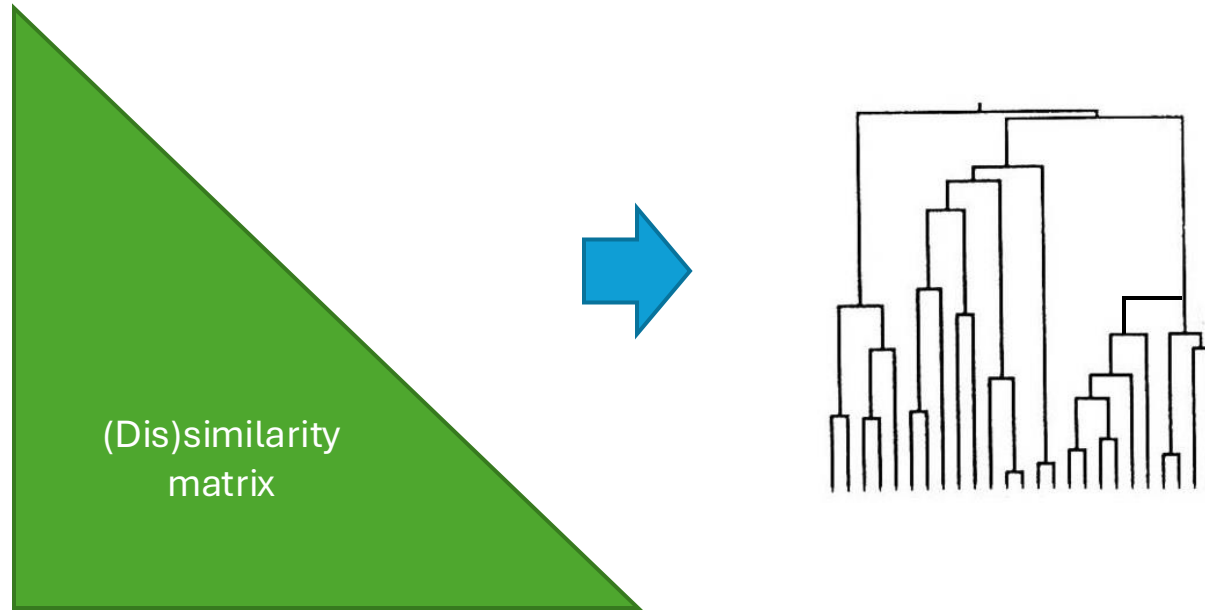
or



Dimension reduction
(e.g. NMDS)

Cluster Analysis

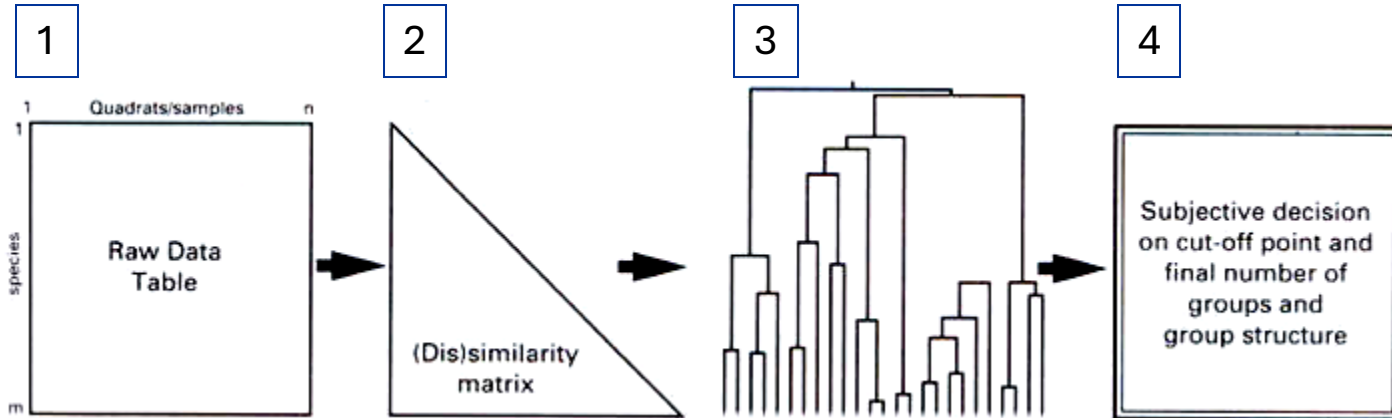
Create a dendrogram by division or agglomeration



Cluster Analysis

Approach:

Raw data [1] to calculate a dissimilarity matrix [2],
visualised in a cluster diagram [3], as baseline for a
grouping structure [4]

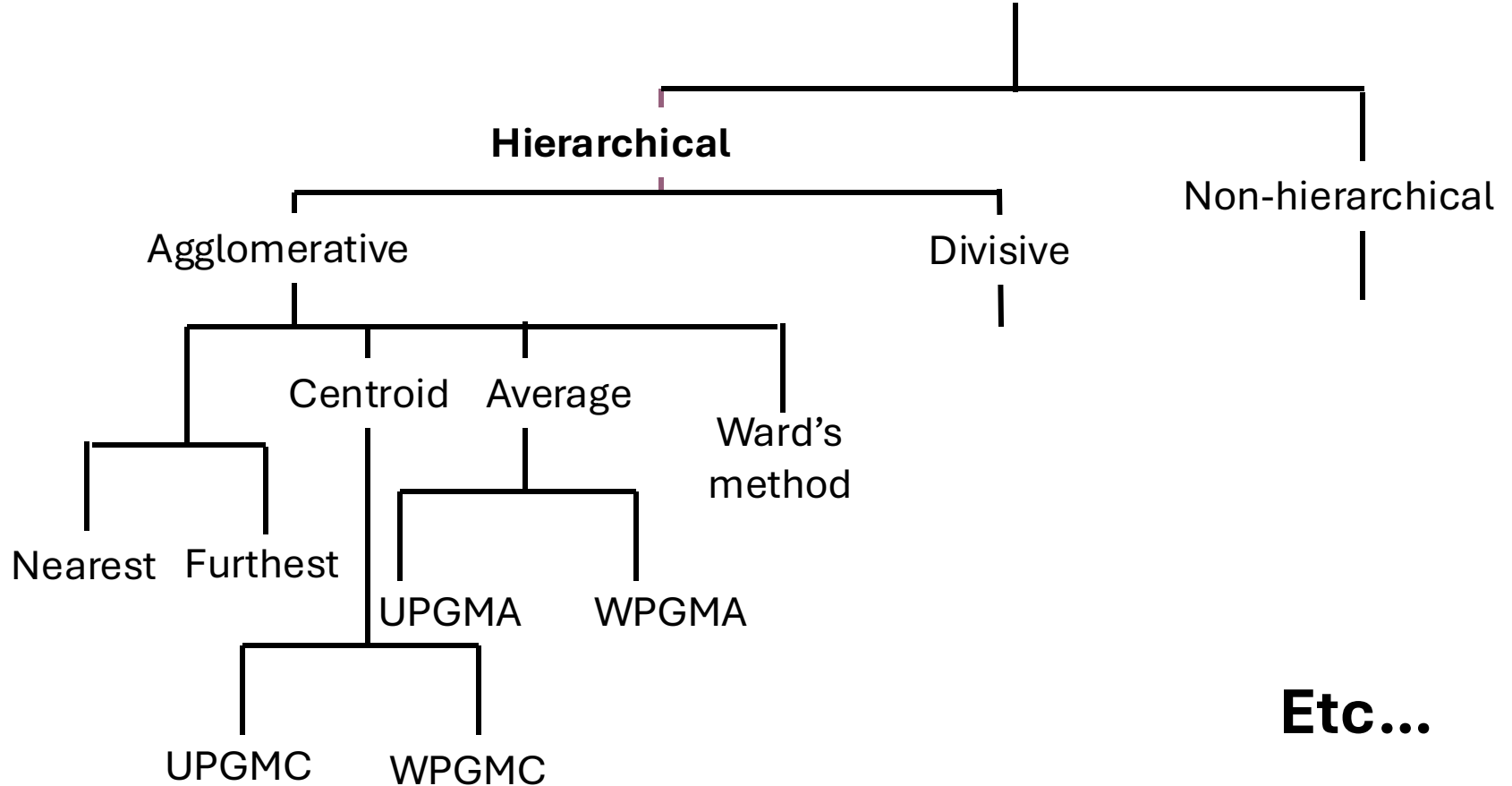


Clustering Algorithms

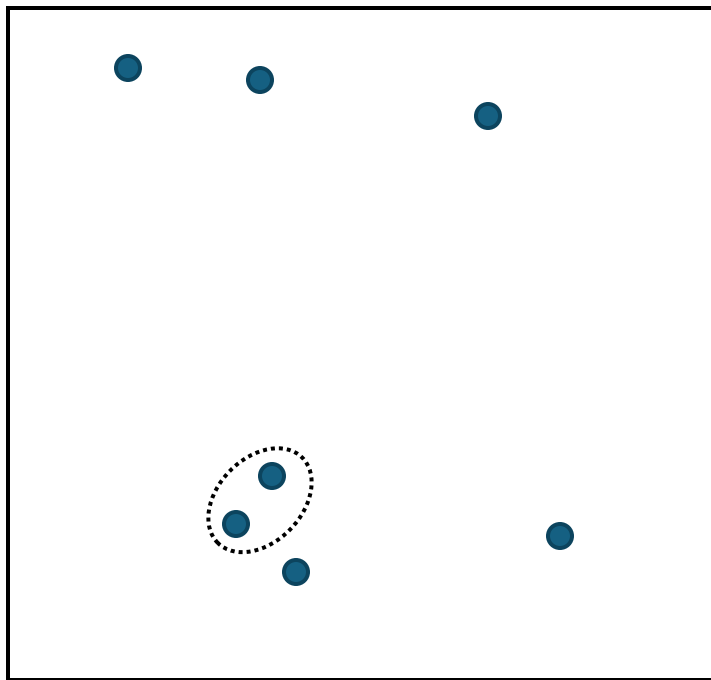
How are groups defined:

- **Hierarchical** algorithms: several smaller groups can be re-joined on a higher level.
- **Non-hierarchical** algorithms: Groups exist without any relation between each other.

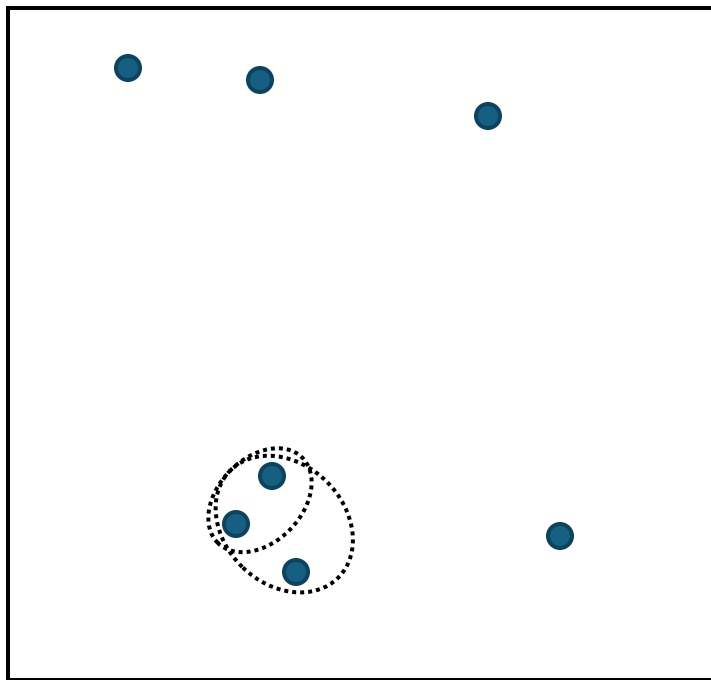
Clustering Algorithms



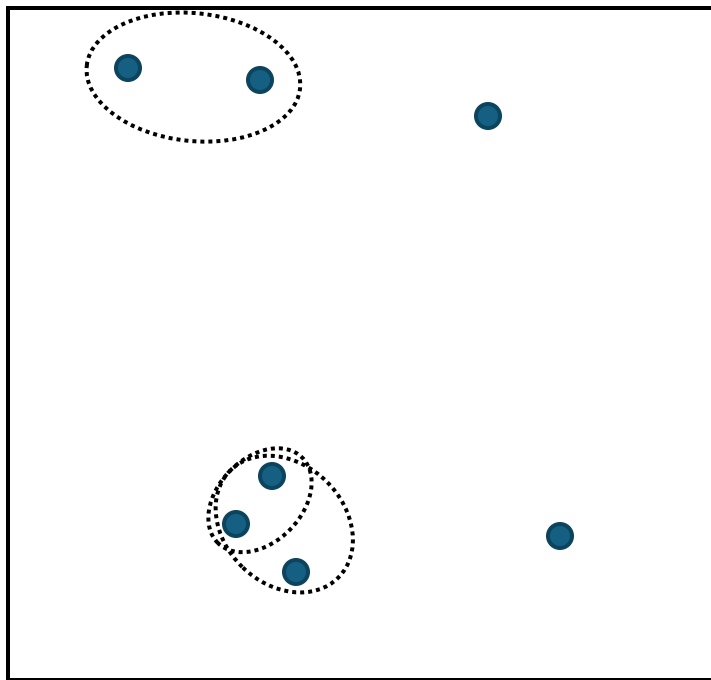
Linkage Algorithms: Single Linkage



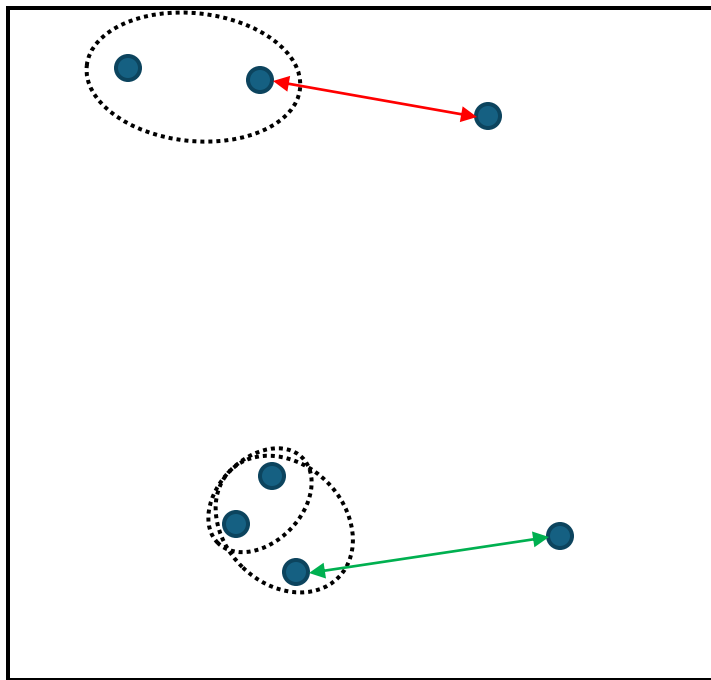
Linkage Algorithms: Single Linkage



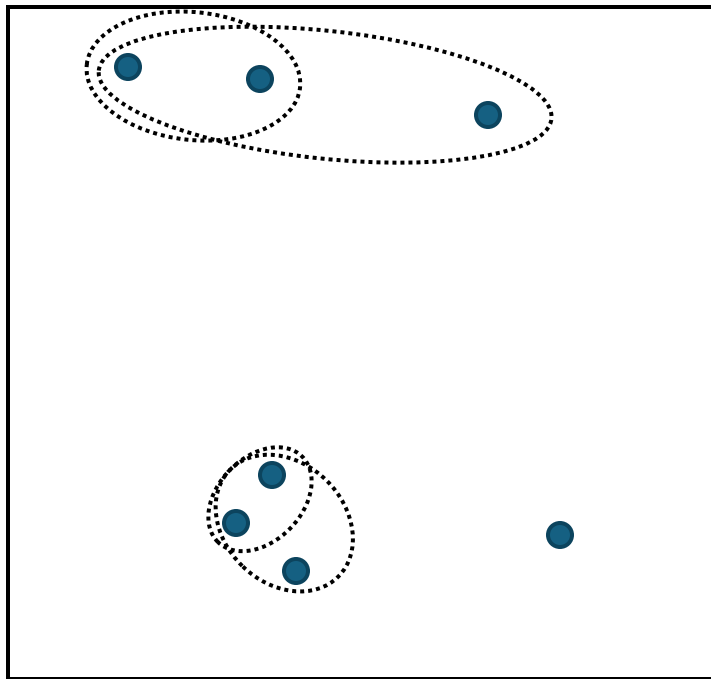
Linkage Algorithms: Single Linkage



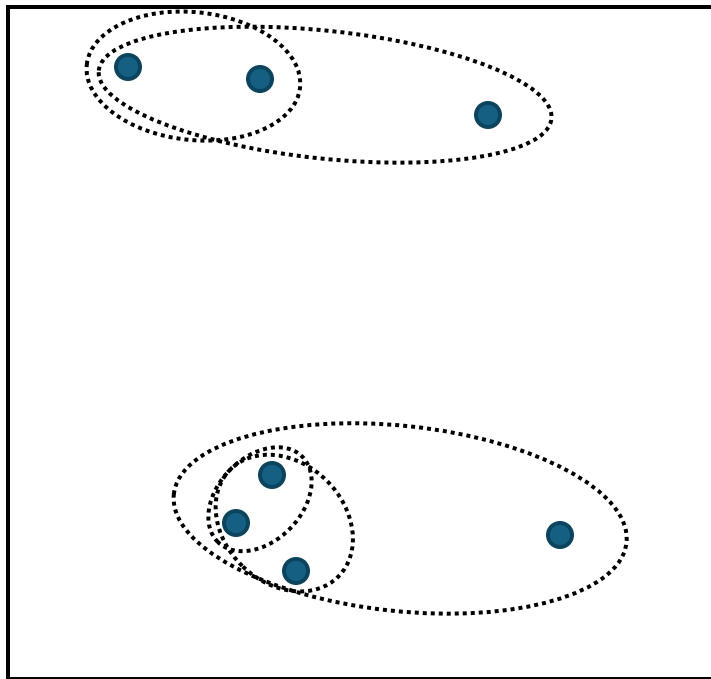
Linkage Algorithms: Single Linkage



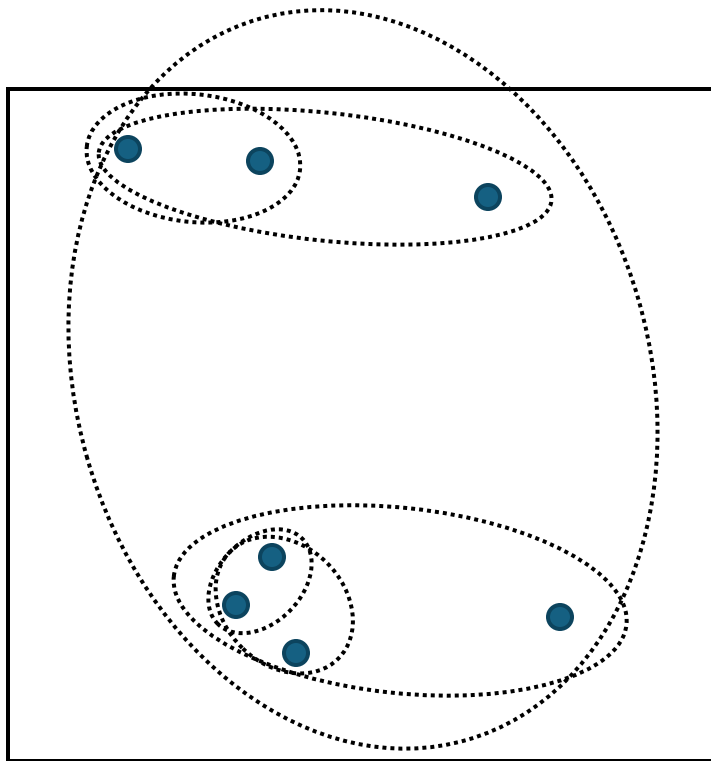
Linkage Algorithms: Single Linkage



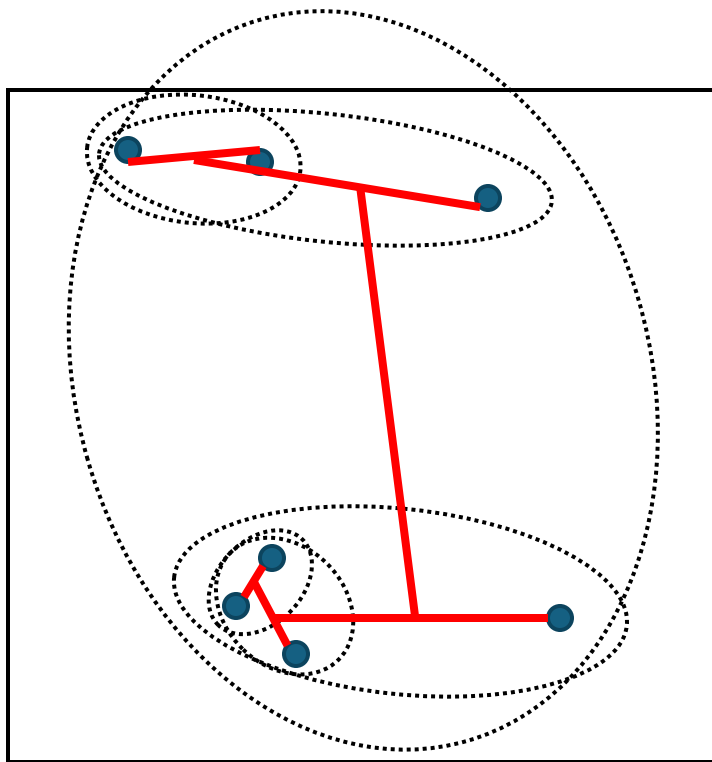
Linkage Algorithms: Single Linkage



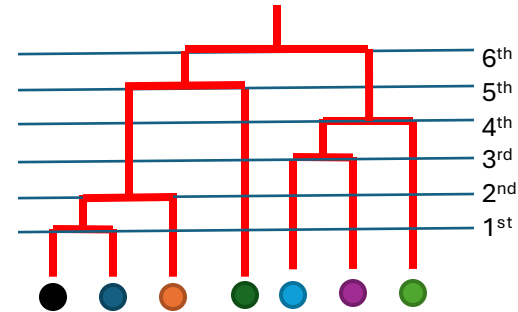
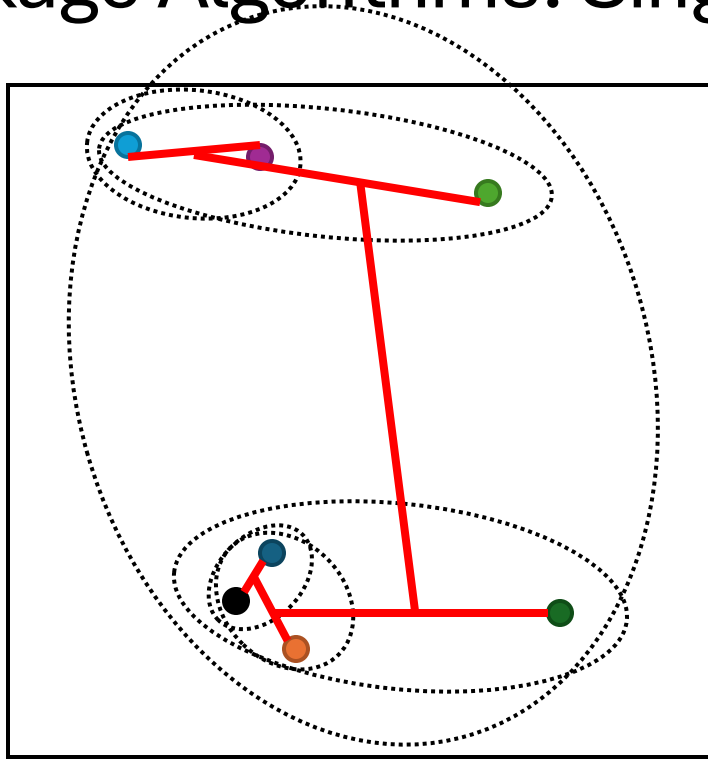
Linkage Algorithms: Single Linkage



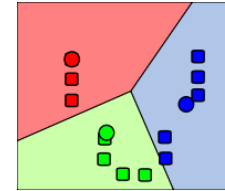
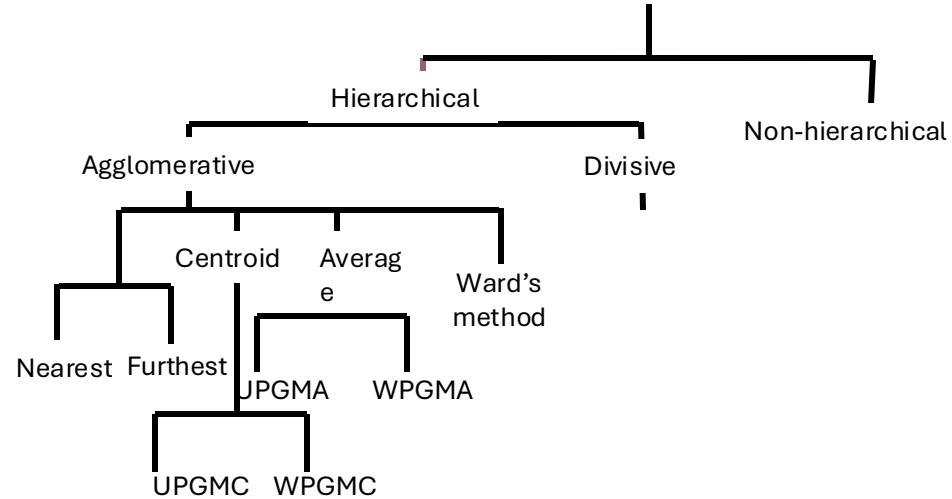
Linkage Algorithms: Single Linkage



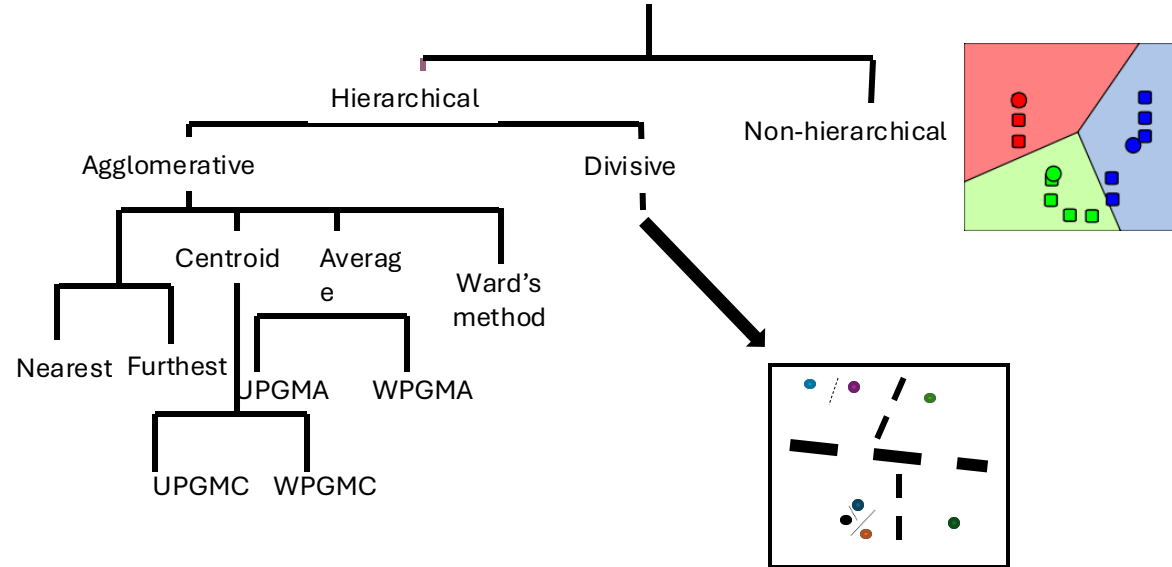
Linkage Algorithms: Single Linkage



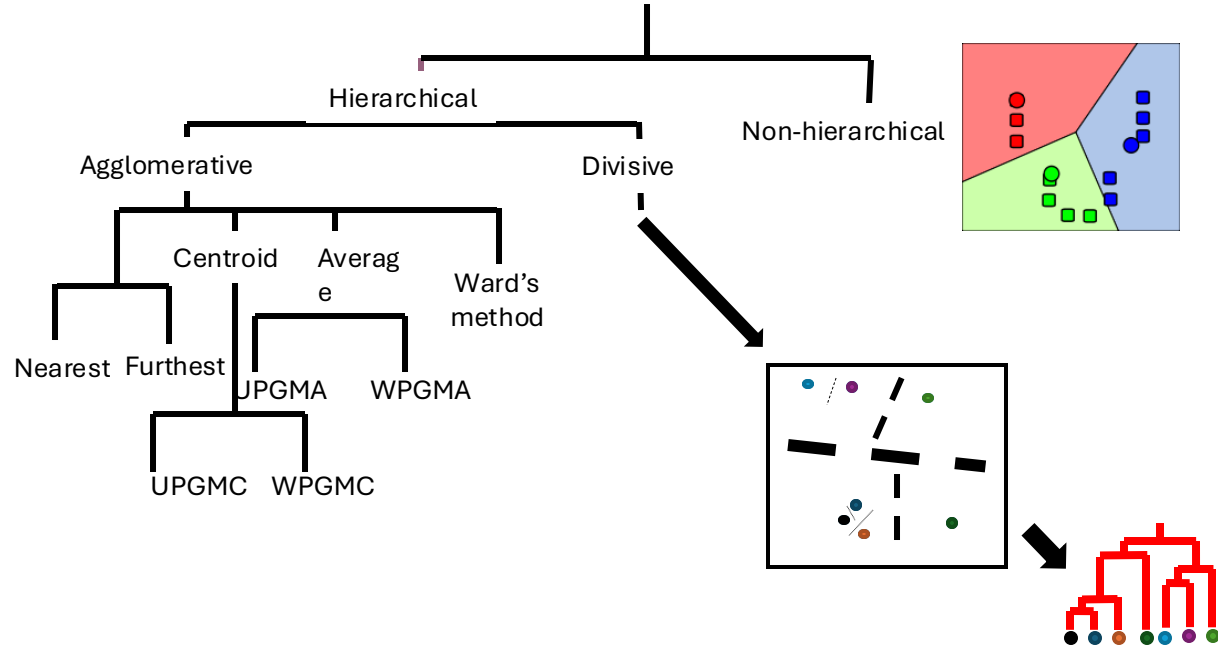
Clustering Algorithms



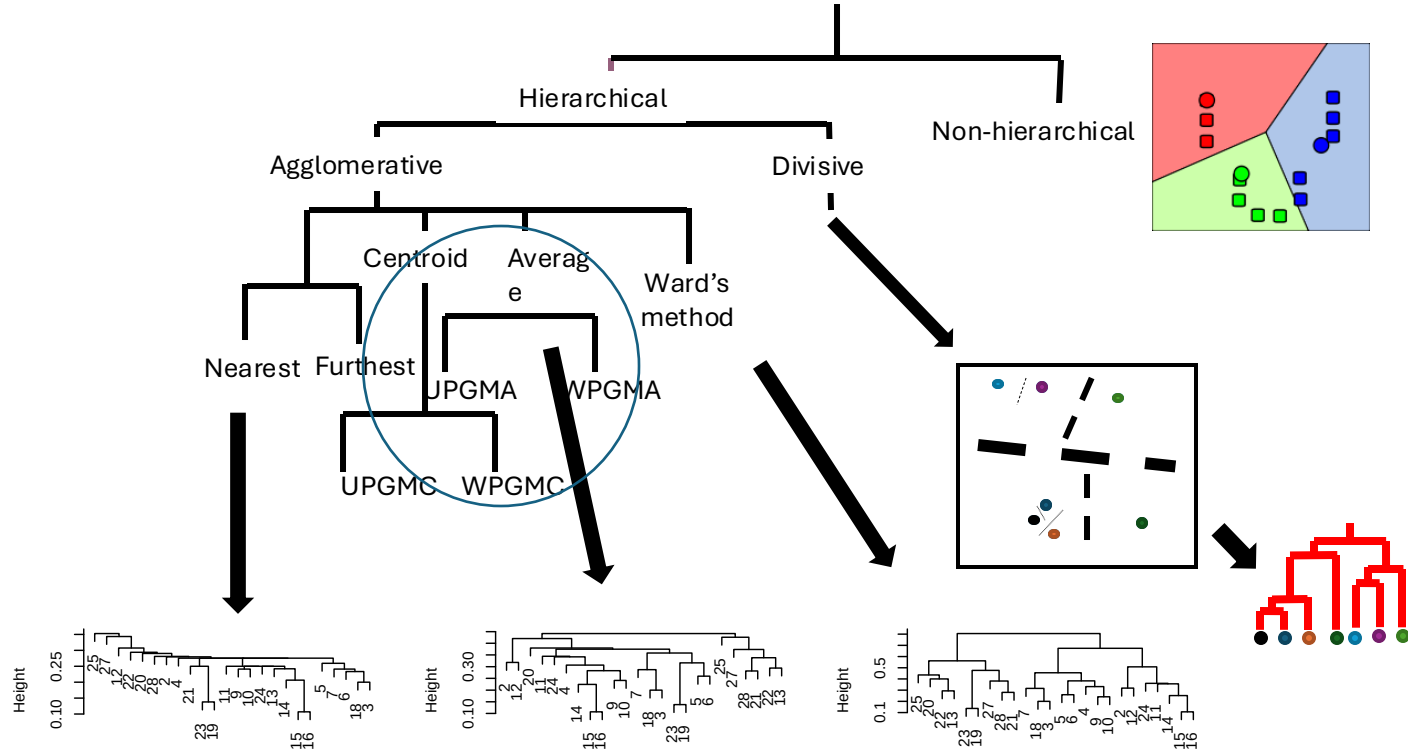
Clustering Algorithms



Clustering Algorithms

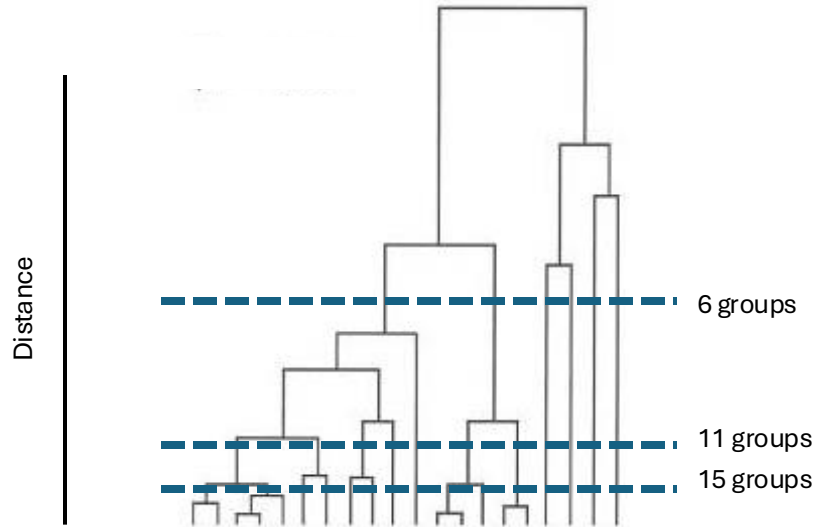


Clustering Algorithms

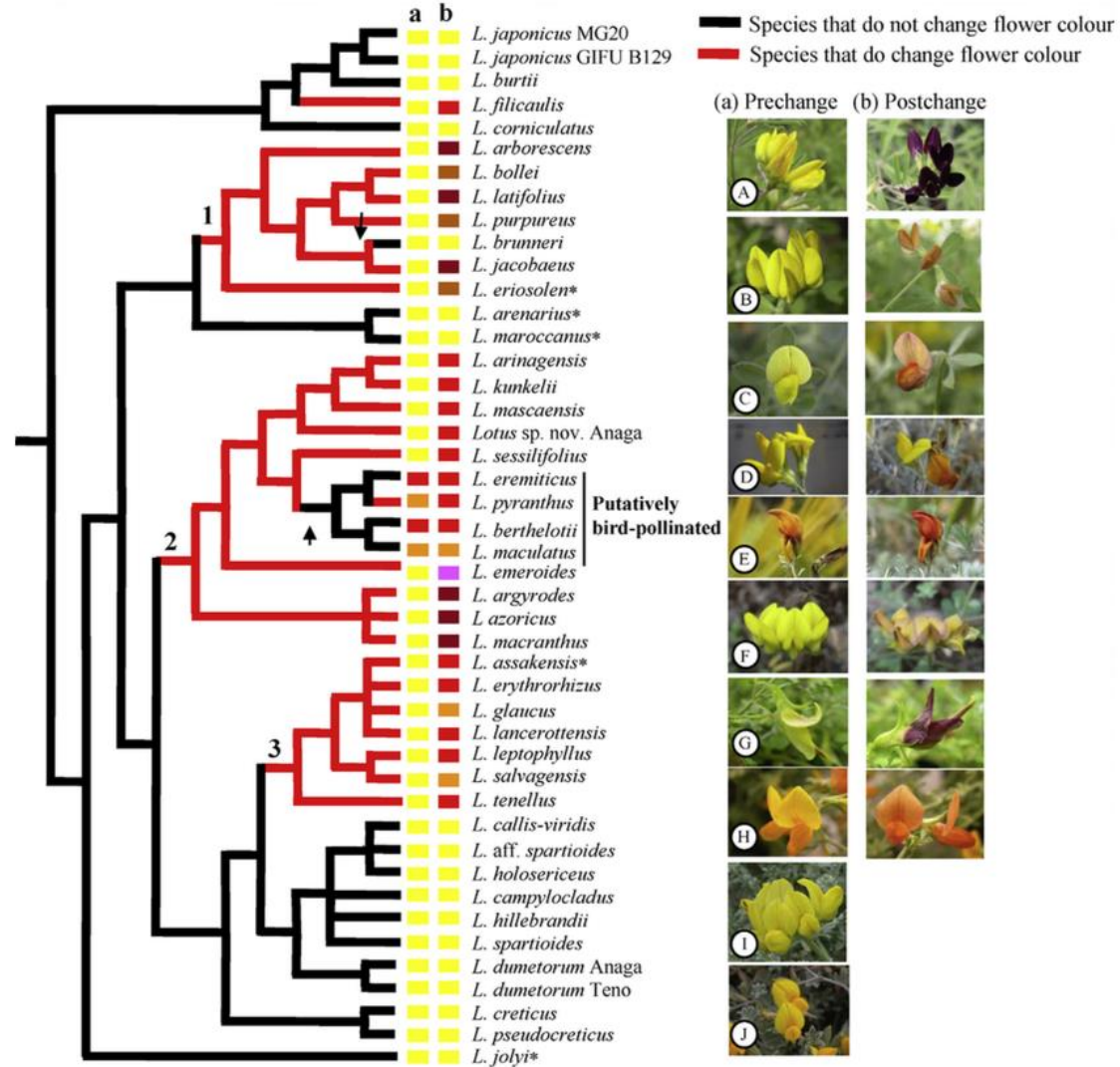


Define groups

Groups can be defined at various levels (*cophenetic distances*)

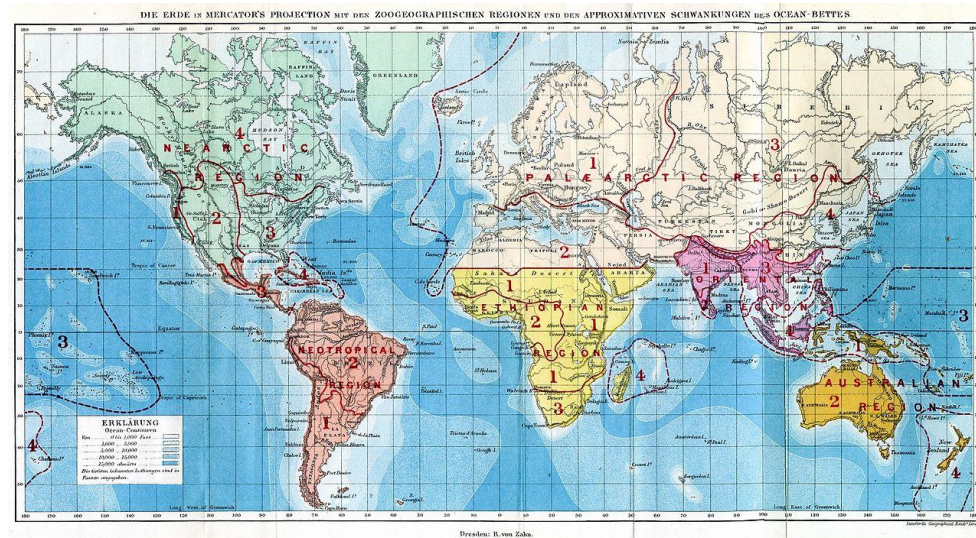


The tree can also be the aim:
Phylogenetic tree of
Lotuses



Mort et al. (2002)
Systematic Botany

Example: Wallace's zoogeographic regions



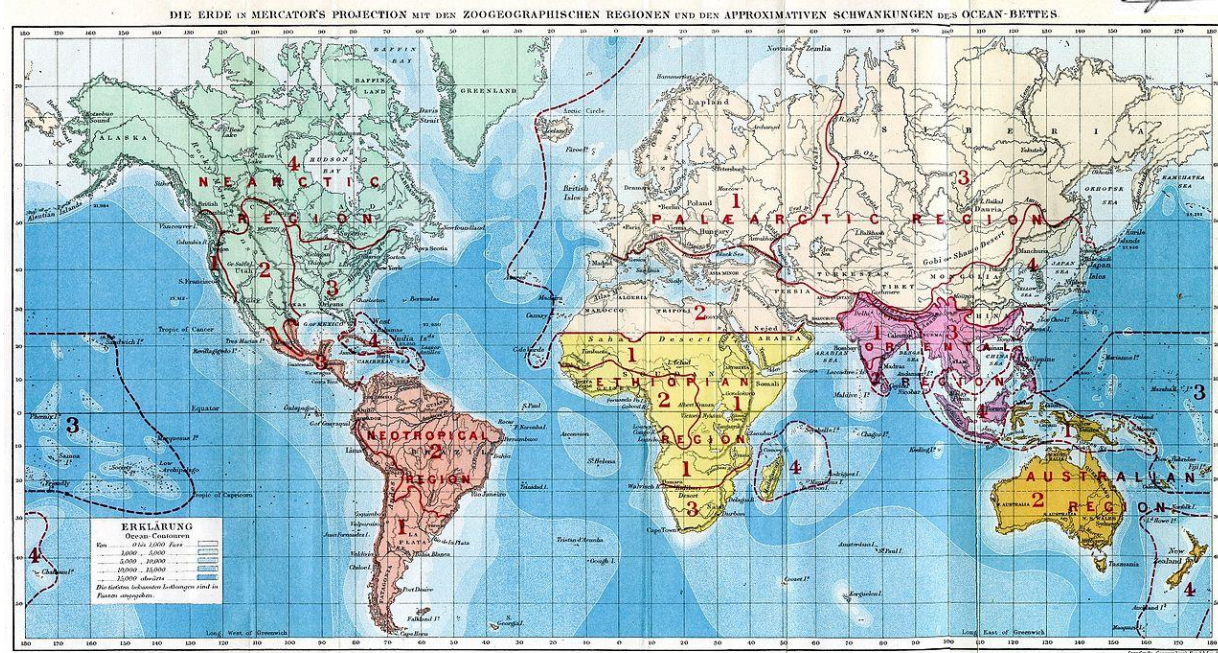
**SPECIAL
PAPER**

A framework for delineating biogeographical regions based on species distributions

Holger Kreft^{1,2*} and Walter Jetz^{2,3}

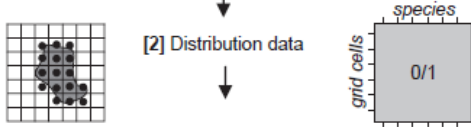


¹Biodiversity, Macroecology and Conservation Biogeography Group, Faculty of Forest Sciences and Forest Ecology, University of Göttingen, Büsgenweg 2, 37077 Göttingen, Germany,
²Ecology, Behavior and Evolution Section, Division of Biological Sciences, University of California San Diego, 9500 Gilman Drive, MC 0116, La Jolla, CA 92093-0116, USA,
³Department of Ecology and Evolutionary Biology, Yale University, PO Box 208106, 165 Prospect Street, New Haven, CT 06520-8106, USA

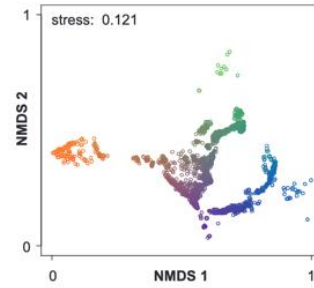


[1] Defining purpose and objectives

[2] Distribution data



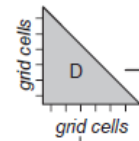
[3] Selection of distance metric



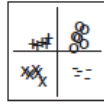
[4] Visualizing turnover



Distance matrix



[5] Ordination

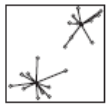


Map of results

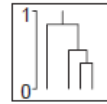


[6] Cluster analysis

Non-hierarchical



Hierarchical



Map of results

Spatial framework for further
historical, ecological, and conservation
biogeographical studies

